Lecture Notes on Symmetry (MTH204)

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Contents

Lecture 1

Course logistics

January 06, 2025

Course Hours

Lecture. Mondays and Wednesdays at 3:00 PM in LH-5.
Tutorials. Fridays at 3:00 PM in LH-5.

Office Hours. Timings to be declared, Venue: AB2-1F7.

Course TAs

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Grading Policy

1. Mid semester exams : 2 x 20 = 40 points.
2. Reading Comprehensions and Quizzes : 10 points.
3. End semester exam : 1 x 40 = 40 points.
4. Attendance: 10 points
   Every student starts with 10 points. Four unexcused absences are permissible without a loss of point. From then onward every unexcused absence will attract a penalty of 1 point per absence.

Syllabus

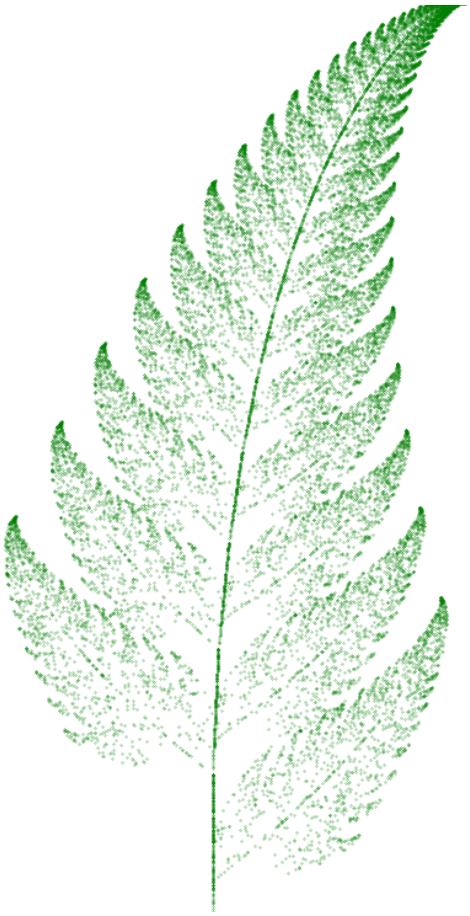
- Definition of symmetry in $\mathbb{R}^2$ and $\mathbb{R}^3$.
- Symmetries of regular planar shapes and their types, composition tables of symmetries.
- Symmetries of tetrahedron and cube, relations between symmetries.
- Axioms of symmetries, groups and subgroups.
- Permutation groups and understanding symmetry through permutation groups, permutation matrices.
- Rotation matrices and reflection matrices, matrix groups.
- Group actions, natural action of the group of symmetries.
- Platonic solids and their symmetries.

Suggested readings

- M. A. Armstrong, *Groups and Symmetry*, Undergraduate Texts in Mathematics, Springer-Verlag, 1988.
- Michael Artin, *Algebra*, Prentice-Hall of India, 1994.

1.1. Thinking of symmetry

Some of the keywords that catch our visual attention while thinking of symmetry are–reflection about a mirror, rotation about an axis, invariance under a transformation, repetition, similarity. We will assign a precise meaning to symmetry in the next lecture. In addition to a mathematical definition of symmetry, it is essential for us to learn how to identify symmetry, how to keep track of multiple symmetries, and most importantly–how to make calculations using symmetry. We will see that our existing understanding of linear algebra and groups would be playing a pivotal role in accomplishing all this. In the mean time, here is a food for thought. Should we say that the following picture (Barnsley's fern) possesses symmetry?



Barnsley's Fern

Lecture 2

Symmetry in $\mathbb{R}^2$ and $\mathbb{R}^3$

January 08, 2025

Let us fix the following notation: $\mathbb{R}^2$ is the plane whose points are characterized by an *arbitrary* choice of origin and x and y coordinates with respect to two non-parallel (and non-antiparallel) directions. Similarly, $\mathbb{R}^3$ is the space whose points are characterized by an *arbitrary* choice of origin and x, y and z coordinates with respect to three non-planar directions. We denote the points of $\mathbb{R}^2$ and $\mathbb{R}^3$ by column vectors. Further, the directions x, y, and z are be taken to be mutually perpendicular. For $v, w \in \mathbb{R}^2$ or $\mathbb{R}^3$, the distance between v and w will be denoted by $d(v, w)$. We will assign more precise meaning to distance in the next lecture.

A subset S of $\mathbb{R}^2$ will be called a *planar shape* and a subset of $\mathbb{R}^3$ will be called a *spatial shape*.

2.1. What is symmetry?

While we make our definition for planar shapes, these hold for spatial shapes as well. Let $S \subseteq \mathbb{R}^2$ be a planar shape. A map $\sigma : \mathbb{R}^2 \to \mathbb{R}^2$ is called a *symmetry* of S if the following hold for σ and S.

(i). σ is distance preserving. That is, for every $v, w \in \mathbb{R}^2$ we have $d(\sigma v, \sigma w) = d(v, w)$.

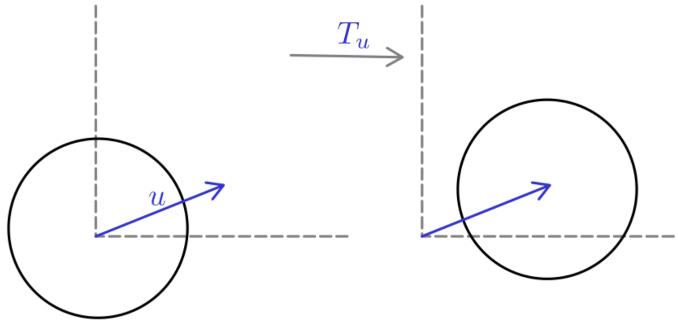
(ii). $\sigma(S) = S$.

Let us see some examples.

1. (*Identity*). The identity map $id : \mathbb{R}^2 \to \mathbb{R}^2$ that maps every $v \in \mathbb{R}^2$ to itself, is a symmetry for every shape S.

2. (*Translation*). Let $u \in \mathbb{R}^2$. The map $T_u : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T_u(v) := v + u$ is a symmetry of $S = \mathbb{R}^2$.

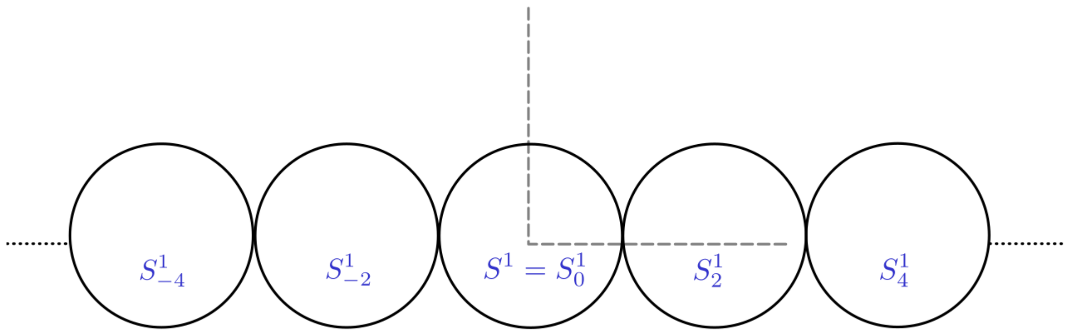
3. Let $S^1 := \{v \in \mathbb{R}^2 : d(v, 0) = 1\}$. This shape is unit circle. Let $0 \neq u \in \mathbb{R}^2$. Is T_u a symmetry of S^1? No. This is because the translation by a nonzero vector displaces the circle and hence $T_u(S^1) \neq S^1$. Thus, the translation T_u, despite being a distance preserving map, is not a symmetry of S^1.



A nontrivial translation T_u is not a symmetry for circle S^1

4. For $n \in \mathbb{Z}$, define
$$S_n^1 := \left\{ \begin{pmatrix} n \\ 0 \end{pmatrix} + v : v \in S^1 \right\}$$
Note that S_n^1 is simply a translation of the unit circle S^1 by $u = \begin{pmatrix} n \\ 0 \end{pmatrix} \in \mathbb{R}^2$.

   Let $S := \bigcup_{n \in \mathbb{Z}} S_{2n}^1$. Can you recognize this shape? It is infinite strip of circles touching each other.



Infinite strip of circles touching each other.

   Note that in this case, $T_u : \mathbb{R}^2 \to \mathbb{R}^2$, where $u = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, is a symmetry of S. Are there other translations that are symmetries of S?

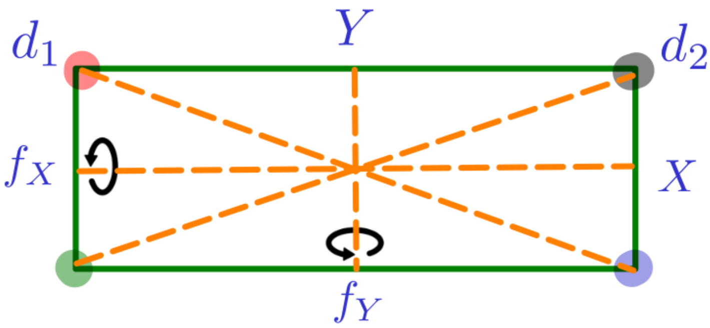
5. With the definition above, what are all symmetries of Barnsley's fern? The only symmetry is the identity map $id : \mathbb{R}^2 \to \mathbb{R}^2$.

Finally, let us make this important observation, which is an immediate consequence of the definition of symmetry.

Remark 2.1 Let $\sigma : \mathbb{R}^2 \to \mathbb{R}^2$ and $\tau : \mathbb{R}^2 \to \mathbb{R}^2$ be symmetries of a shape S. Then the compositions $\sigma\tau : \mathbb{R}^2 \to \mathbb{R}^2$ and $\tau\sigma : \mathbb{R}^2 \to \mathbb{R}^2$ are also symmetries of S.

2.2. Symmetries of a rectangle

Now, we are going to observe symmetries in a rectangle and tabulate them. In the rectangular figure below, we have marked X and Y axes, and the two diagonals d_1 and d_2.



We observe following symmetries in rectangle.

(i). The identity map $id : \mathbb{R}^2 \to \mathbb{R}^2$.

(ii). The flipping f_X about X-axis. Observe that this flipping map $f_X : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by

$$f_X \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ -b \end{pmatrix}$$

(iii). The flipping f_Y about Y-axis. This flipping map $f_Y : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by

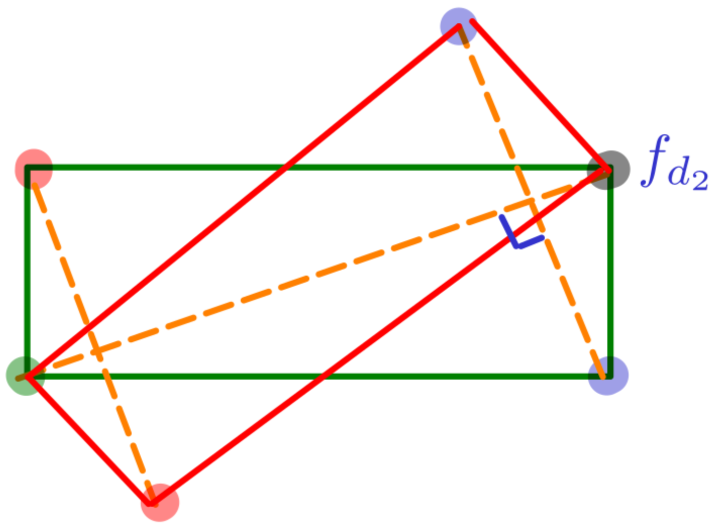
$$f_Y \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ b \end{pmatrix}$$

(iv). What about other symmetries of a rectangle? The composition map $f_X f_Y : \mathbb{R}^2 \to \mathbb{R}^2$ is given by

$$f_X f_Y \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ -b \end{pmatrix}$$

This is actually the rotation map that rotates a point by the angle π (in anticlockwise direction). We denote it by r_π. What about $f_Y f_X$? We note that in this case $f_Y f_X = f_X f_Y = r_\pi$. That the two symmetries f_X and f_Y commute in this case is incidental. In general the compositions $\sigma\tau$ and $\tau\sigma$ of two symmetries σ and τ need not be equal.

(v). Let us try to observe more symmetries? What about flipping along a diagonal?

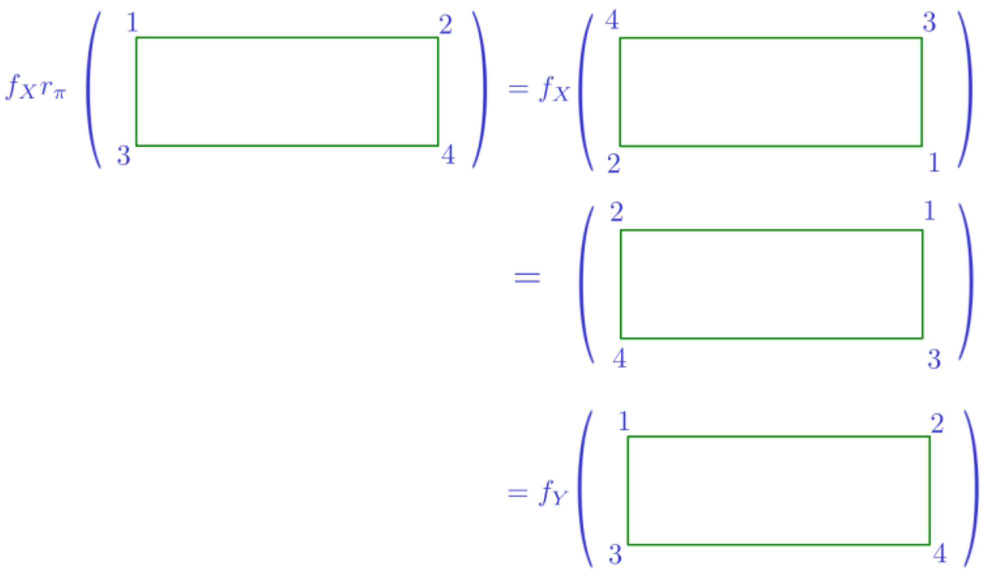


Note that the flipping along diagonal d_2 takes our (green) rectangle to another (red) rectangle. Thus f_{d_2} is not a symmetry, unless our rectangle is a square. Similarly, flipping along the other diagonal is also not a symmetry of the rectangle.

(vi). What about composition $f_X r_\pi$? We note that

$$f_X r_\pi \begin{pmatrix} a \\ b \end{pmatrix} = f_X \begin{pmatrix} -a \\ -b \end{pmatrix} = \begin{pmatrix} -a \\ b \end{pmatrix} = f_Y \begin{pmatrix} a \\ b \end{pmatrix}$$

Hence $f_X r_\pi = f_Y$, and the composition did not yield a new symmetry. Instead of tracking coordinates to recognize symmetries we can just track the movement of four vertices and conclude that $f_X r_\pi = f_Y$.



A rectangle has only four symmetries: id, f_X, f_Y, r_π. After a few lectures we will be able to convince ourselves that there are no more symmetries of a rectangle . For notational convenience, we denote id by 1 and consider the set of symmetries of the rectangle.

$$\mathrm{sym}(R) := \{1, f_X, f_Y, r_\pi\}$$

The following is the composition table of $\mathrm{sym}(R)$.

	1	f_X	f_Y	r_π
1	1	f_X	f_Y	r_π
f_X	f_X	1	r_π	f_Y
f_Y	f_Y	r_π	1	f_X
r_π	r_π	f_Y	f_X	1

The entries highlighted blue indicate that $f_X r_\pi = f_Y$. Thus, in this composition table, the header column signifies the symmetry operation to be performed first and the header row signifies the symmetry operation to be performed next in order to get the symmetry at corresponding entry in the composition table. Incidentally, in this table we have $ab = ba$ for every $a, b \in \mathrm{sym}(R)$. Hence, this composition table is *commutative* or *abelian*.

2.3. Symmetries of an equilateral triangle

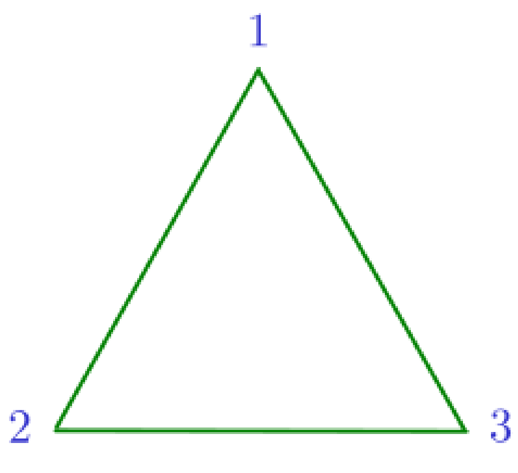
An equilateral triangle possesses the following symmetries.

(i). 1: the identity symmetry.

(ii). $r_{2\pi/3}$: rotation by the angle $2\pi/3$ in anti-clockwise direction about the centre of the triangle.

(iii). $r_{4\pi/3}$: rotation by $4\pi/3$ in anti-clockwise direction about the centre of the triangle.

(iv). f_1, f_2, f_3: where f_i denotes the flipping about median of the triangle passing through vertex i, where i is *as per the labeling in the initial configuration*.



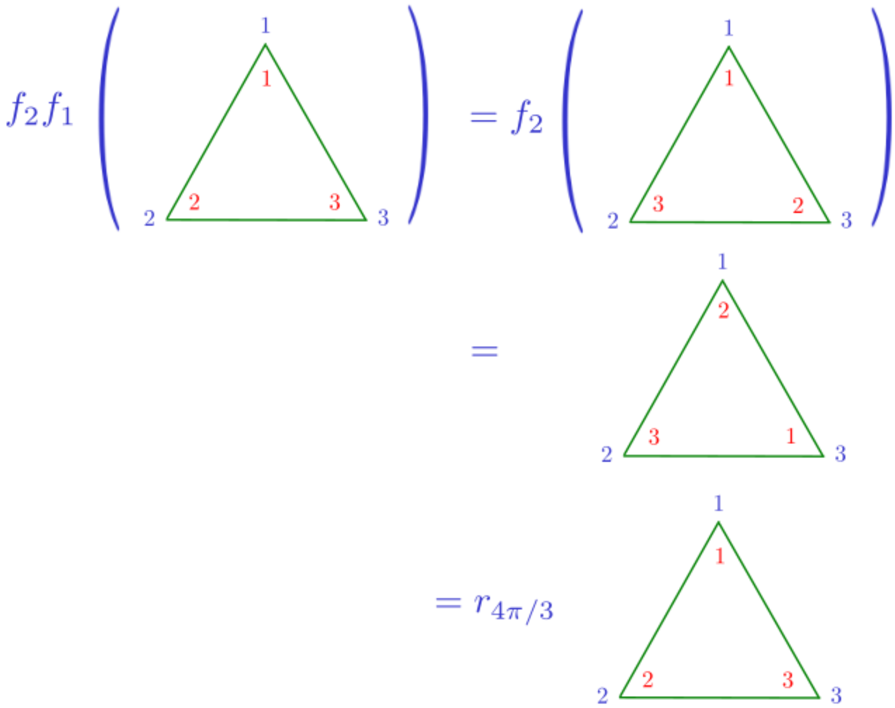
Thus, the set of symmetries of an equilateral triangle Δ is

$$\text{sym}(\Delta) = \{1, r_{2\pi/3}, r_{4\pi/3}, f_1, f_2, f_3\}$$

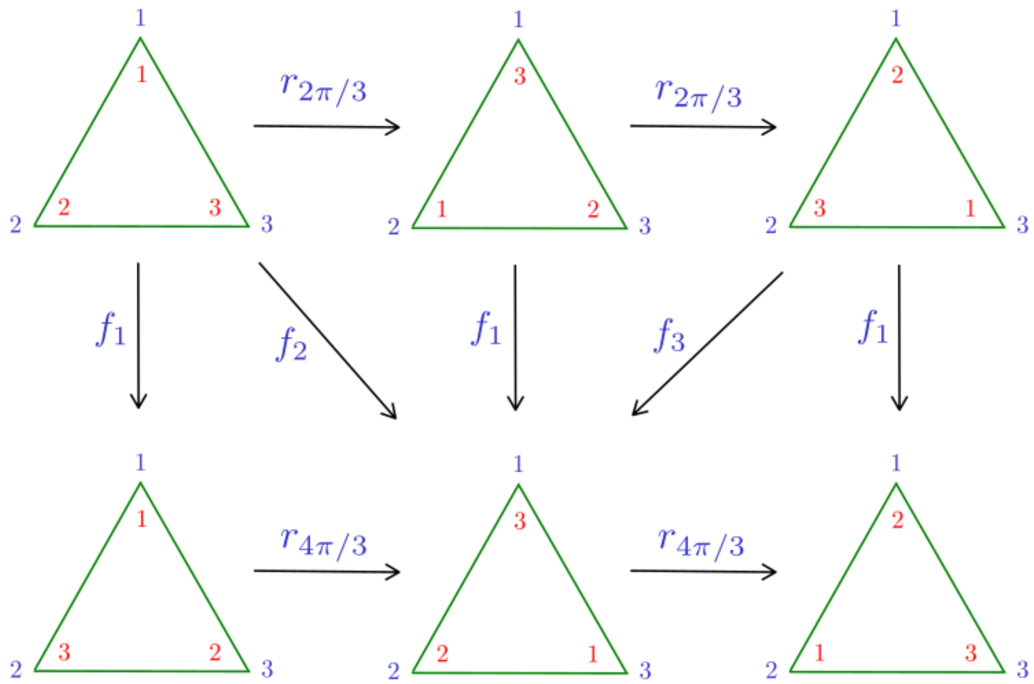
You should confirm the following composition table of $\text{sym}(\Delta)$.

	1	$r_{2\pi/3}$	$r_{4\pi/3}$	f_1	f_2	f_3
1	1	$r_{2\pi/3}$	$r_{4\pi/3}$	f_1	f_2	f_3
$r_{2\pi/3}$	$r_{2\pi/3}$	$r_{4\pi/3}$	1	f_2	f_3	f_1
$r_{4\pi/3}$	$r_{4\pi/3}$	1	$r_{2\pi/3}$	f_3	f_1	f_2
f_1	f_1	f_3	f_2	1	$r_{4\pi/3}$	$r_{2\pi/3}$
f_2	f_2	f_1	f_3	$r_{2\pi/3}$	1	$r_{4\pi/3}$
f_3	f_3	f_2	f_1	$r_{4\pi/3}$	$r_{2\pi/3}$	1

Here, the entries highlighted blue signify the composition $f_2 f_1 = r_{4\pi/3}$. To see this, we may go through the following sequence of transformations. In the figure below, the initial configuration is labeled blue and the movement of vertices is tracked by the labels colored red.



The six symmetric movements of an equilateral triangle and their relations can be read through the following picture. It is a good task to read this picture together with the composition table of $\text{sym}(\Delta)$.



2.4. What is common between the two composition tables?

Recall that if $a, b \in \mathrm{sym}(R)$ then $ab = ba$. However, in contrast, there are $a, b \in \mathrm{sym}(\Delta)$ such that $ab \neq ba$. For example, $f_2 f_1 = r_{4\pi/3}$, but $f_1 f_2 = r_{2\pi/3}$. Despite this, the composition tables for $\mathrm{sym}(R)$ and $\mathrm{sym}(\Delta)$ have the following common features.

(i). If $a, b, c \in \mathrm{sym}(\cdot)$, then $(ab)c = a(bc)$.

(ii). For every $a \in \mathrm{sym}(\cdot)$, there is some $a' \in \mathrm{sym}(\cdot)$ such that $aa' = a'a = 1$.

You should verify these. In the next lecture, we will axiomatize these features to say that symmetries of a shape form a group.

Lecture 3

Groups: the storekeepers of symmetries

January 10, 2025

In the last lecture, we observed that the composition on the set of symmetries is associative and that every symmetry comes equipped with an opposite symmetry. These features will be axiomatized to the concept of groups. At this point, you are encouraged to revise Lecture 18 (October 26, 2023) of your MTH101 (Basic Linear Algebra) course.

3.1. Groups

A *group* consists of two mathematical objects — first one - a set X, and second one - a binary operation $* : X \times X \to X$ that satisfies the following properties.

I. **Associativity**.
$$(a * b) * c = a * (b * c)$$
for every triplet a, b, c in X. Here, the notation $a * b$ stands for the image of $(a, b) \in X \times X$ under the binary operation $*$.

II. **Existence of a neutral element**. There exists an element $1 \in X$ such that for every $a \in X$,
$$\begin{aligned} & a * 1 = a, \\ \text{and,} \quad & 1 * a = a \end{aligned}$$

III. **Existence of inverse**. For every $a \in X$, there exists $a' \in X$ such that
$$\begin{aligned} & a * a' = 1, \\ \text{and,} \quad & a' * a = 1 \end{aligned}$$

For simplicity of notation, we write: $ab := a*b$. A group $(X, *)$ is said to be a *commutative group* or an *abelian group*, if for every $a, b \in X$ we have $ab = ba$. Can a group have two distinct neutral elements, say 1 and $1'$? No. The following computation clears this point.
$$1 = 1 * 1' = 1'$$

Here the first equality holds because $1'$ is a neutral element, and the second equality holds because 1 is a neutral element. Thus, a group has *unique* neutral element.

The same question for inverses — *Can an element $a \in X$ have two distinct inverses a' and a''?* No, again, as the following computation shows.

$$a' = a' * 1 = a' * (a * a'') = (a' * a) * a'' = 1 * a'' = a''$$

We attribute

- first and last equalities to the fact that 1 is neutral element.
- the second equality to the fact that a'' is inverse of a.
- the third equality to the associativity property of groups.
- the fourth equality to the fact that a' is inverse of a.

Thus, uniqueness of neutral element and inverses is not a part of group axioms, but rather a consequence of these. When the binary operation is clear from the context, then we quietly drop $*$ from the notation $(X, *)$ and say that X is a group.

Number of elements in X is called the *order* of the group X. Groups that have infinitely many elements are called *infinite groups*.

3.2. Examples and non-examples of groups

1. First and the foremost, symmetry groups of shapes are examples of groups. Thus, $\mathrm{sym}(R)$ is an abelian group of order four. Interestingly, in this group every element is inverse of itself! That is, $a = a'$ for every $a \in \mathrm{sym}(R)$.

   The group $\mathrm{sym}(\Delta)$ is a nonabelian group of order six. The three flippings f_1, f_2 and f_3, and the neutral element 1 are their own inverses, and two rotations are inverses of each other.

   The group $\mathrm{sym}(\mathbb{R}^2)$ is an infinite nonabelian group. This group is nonabelian because rotation and translation maps do not commute, in general. The inverse of translation T_u is the translation T_{-u}.

2. $(\mathbb{Z}, *)$. *integers under the binary operation of multiplication* **do not** form a group. While the integer 1 is a neutral element for multiplication, there are elements in $\mathbb{Z}$, such as 0 and 2, that do not have inverse. In fact, no integer except ± 1 has multiplicative inverse within integers.

3. $(\mathbb{Q}, +)$. *rational numbers under the binary operation of addition* form a group.

4. Consider the *set of clock hours*: $X := \{12, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$, and the binary operation of *addition of clock hours*. Thus, $10 + 3 = 1$, and $12 + 7 = 7$ under this binary operation. It forms a group, and the neural element is 12. The inverse of 8 is 4, and the inverse of 1 is 11.

5. Consider the set $X := \{1, 3, 7, 9\}$. For $a, b \in X$, define $a * b$ to be the *rightmost digit in the usual multiplication* of a and b. It is interesting to observe that it forms a group! In this group, 1 is the neutral element, 3 and 7 are inverses of each other, and 9 is its own inverse.

6. Consider a set $X := \{a, b, c\}$ and put a binary operation on it through the following table.

$*$	a	b	c
a	b	a	c
b	c	b	a
c	a	c	b

   This $b * c = c$ and $c * a = c$. Does this binary operation define a group? No! The operation is neither associative, nor does it have a neutral element. Existence of inverse is, of course, out of question!

7. Consider the set X that consists of 2×2 matrices whose entries are integers and determinant is nonzero. On X, consider the binary operation of matrix multiplication. Does it form a group? Well, while there is indeed a neutral element, namely the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in X$, there are elements such as $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \in X$, which do not have inverses. Thus, X is not a group under matrix multiplication.

8. *Orthogonal matrices.* Consider the set $\mathrm{O}_n(\mathbb{R})$ consisting of $n \times n$ invertible matrices with entries in $\mathbb{R}$ such that $A^t = A^{-1}$. Here, A^t denotes the transpose of A, and A^{-1} is the multiplicative inverse of A. Such matrices are called *orthogonal matrices*. On this set, consider the binary operation of matrix multiplication? Do we have a group?

   First of all, if $A, B \in \mathrm{O}_n(\mathbb{R})$, then AB is invertible (being the product of two invertible matrices). Further, $(AB)^t = B^t A^t = B^{-1}A^{-1} = (AB)^{-1}$. Thus, AB indeed belongs to $\mathrm{O}_n(\mathbb{R})$. Also, if $A \in \mathrm{O}_n(\mathbb{R})$, then

$$(A^{-1})^t = (A^t)^t = A = (A^{-1})^{-1}$$

   and thus, $A \in \mathrm{O}_n(\mathbb{R})$. The identity matrix I_n also belongs to $\mathrm{O}_n(\mathbb{R})$ and $A \in \mathrm{O}_n(\mathbb{R})$ inherits associativity from the associativity of multiplication of matrices. Thus, the set $\mathrm{O}_n(\mathbb{R})$ of orthogonal matrices forms a group under matrix multiplication.

9. This one is a curious exercise. Consider the set

$$X := \left\{ \begin{pmatrix} a & a \\ a & a \end{pmatrix} : a \in \mathbb{Q}^\times \right\}$$

   On this set, consider the binary operation of matrix multiplication. Show that X is a group under this binary operation. The neutral element of this group is the matrix $\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \in X$.

In the following lectures, our task will be to understand the group $\mathrm{sym}(\mathbb{R}^2)$ and utilize it to understand symmetry of various shapes.

Lecture 4

Pinpointing symmetries in $\mathbb{R}^2$

January 13, 2025

We have seen that translations, rotations and reflections are symmetries. We know that their compositions are also symmetries. In this lecture, we aim to classify all distance preserving maps $\sigma : \mathbb{R}^2 \to \mathbb{R}^2$. The central role in this classification is played by the *dot product* that helps us measure distances and angles.

4.1. Measuring distances and angles in $\mathbb{R}^2$

Before we talk about preserving distances, we need to equip ourselves with the tools that help us measure distances. Let us systematically recall and develop these.

(i). *Dot product*. We define the map $\langle \ \cdot \ \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ by

$$\langle v, w \rangle := v^t w, \qquad \text{where } v, w \in \mathbb{R}^2$$

Thus, if $v = \begin{pmatrix} a \\ b \end{pmatrix}$ and $w = \begin{pmatrix} c \\ d \end{pmatrix}$, then $\langle v, w \rangle = ac + bd$. This is called the *dot product* of v and w. Dot product is an example of *symmetric bilinear map*, in the sense that for every $v, w, v', w' \in \mathbb{R}^2$ and for every $\lambda \in \mathbb{R}$, we have the following.

- $\langle v, w \rangle = \langle w, v \rangle$.
- $\langle v + \lambda v', w \rangle = \langle v, w \rangle + \lambda \langle v', w \rangle$.
- $\langle v, w + \lambda w' \rangle = \langle v, w \rangle + \lambda \langle v, w' \rangle$.

(ii). *Length*. We first note that for every $v \in \mathbb{R}^2$, we have $\langle v, v \rangle \geq 0$, and the equality holds if and only if $v = 0$. The *length* is the function $\ell : \mathbb{R}^2 \to \mathbb{R}^+ \cup \{0\}$ given by

$$\ell(v) := \sqrt{\langle v, v \rangle}$$

Thus, if $v = \begin{pmatrix} a \\ b \end{pmatrix}$ then $\ell(v) = (a^2 + b^2)^{1/2}$. We observe that $\ell(v) = \ell(-v)$.

(iii). *Distance*. The *distance* is the function $d : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^+ \cup \{0\}$ given by

$$d(v, w) := \ell(v - w)$$

It has the following properties.

- *Triangle inequality.* $d(v_1, v_2) + d(v_2, v_3) \geq d(v_1, v_3)$ for every $v_1, v_2, v_3 \in \mathbb{R}^2$.
- *Symmetry.* $d(v, w) = d(w, v)$ for every $v, w \in \mathbb{R}^2$.

(iv). *Angle cosine.* *The angle cosine* is the function $\theta : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ given by

$$\theta(v, w) := \frac{\langle v, w \rangle}{\ell(v)\ell(w)}$$

Thus, if v and w are mutually orthogonal then $\theta(v, w) = 0$; $\theta(v, v) = 1$; and $\theta(v, -v) = -1$.

(v). *Dot product in terms of length.* For $v, w \in \mathbb{R}^2$, we perform the following calculation.

$$\begin{aligned} \ell(v - w)^2 &= \langle v - w, v - w \rangle \\ &= \langle v, v \rangle - \langle v, w \rangle - \langle w, v \rangle + \langle w, w \rangle \\ &= \ell(v)^2 - 2\langle v, w \rangle + \ell(w)^2 \end{aligned}$$

Therefore,

$$\langle v, w \rangle = \frac{1}{2} \left(\ell(v)^2 + \ell(w)^2 - \ell(v - w)^2 \right)$$

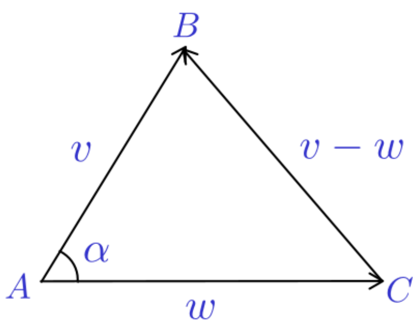
We replace $\langle v, w \rangle$ by $\theta(v, w)\ell(v)\ell(w)$, using the definition of angle cosine, and obtain

$$\theta(v, w)\ell(v)\ell(w) = \frac{1}{2} \left(\ell(v)^2 + \ell(w)^2 - \ell(v - w)^2 \right)$$

Thus,

$$\theta(v, w) = \frac{\ell(v)^2 + \ell(w)^2 - \ell(v - w)^2}{2\ell(v)\ell(w)}$$

This is nothing but the following cosine formula that we saw in our school.



$$\cos \alpha = \frac{|AB|^2 + |AC|^2 - |BC|^2}{2|AB||AC|}$$

Thus, the nomenclature in our definition of angle cosine is justified.

We are now prepared to classify symmetries in $\mathbb{R}^2$.

4.2. T_u and $O_2(\mathbb{R})$ is all that it takes

Theorem 4.1. *Let $\sigma : \mathbb{R}^2 \to \mathbb{R}^2$ be a distance preserving map. Then there exists $u \in \mathbb{R}^2$ and an orthogonal matrix $A \in O_2(\mathbb{R})$ such that $\sigma = T_u \circ A$. Therefore, every distance preserving map in $\mathbb{R}^2$ is obtained by composing multiplication by an orthogonal matrix and translation.*

Proof. Let $u = \sigma(0) \in \mathbb{R}^2$ and define $\sigma' : \mathbb{R}^2 \to \mathbb{R}^2$ by $\sigma'(v) = \sigma(v) - u$. Thus, $\sigma' = T_{-u}\sigma$. We make a number of observations about σ'.

(i). *$\sigma'(0) = 0$ and $\sigma' : \mathbb{R}^2 \to \mathbb{R}^2$ is distance preserving.*
This is an easy check. First, $\sigma'(0) = \sigma(0) - u = \sigma(0) - \sigma(0) = 0 \in \mathbb{R}^2$. Further, for $v, w \in \mathbb{R}^2$,

$$\begin{aligned} d(\sigma' v, \sigma' w) &= \ell(\sigma' v - \sigma' w) \\ &= \ell(\sigma v - \sigma(0) - \sigma w + \sigma(0)) \\ &= \ell(\sigma v - \sigma w) \\ &= d(\sigma v, \sigma w) \\ &= d(v, w) \end{aligned}$$

where the last equality holds due to the fact that σ is distance preserving.

(ii) As a consequence of (i), the map σ' is length preserving as well. This is because

$$\begin{aligned} \ell(\sigma' v) &= d(\sigma' v, 0) & \\ &= d(\sigma' v, \sigma'(0)) & \text{since } \sigma'(0) = 0 \\ &= d(v, 0) & \text{since } \sigma' \text{ is distance preserving} \\ &= \ell(v - 0) = \ell(v) & \end{aligned}$$

(iii). *σ' preserves the dot product and angles.*
It is enough to show that $\langle \sigma' v, \sigma' w \rangle = \langle v, w \rangle$. Since

$$\langle v, w \rangle = \frac{1}{2}\left(\ell(v)^2 + \ell(w)^2 - \ell(v - w)^2\right)$$

and ℓ is preserved by σ', it follows that the dot product is also preserved by σ'. Further, since θ has expression in terms of dot product and ℓ, which are preserved by σ', it follows that angles are preserved by σ'.

(iv). *σ' is linear.*
This is most surprising claim. We are required to show that for every $v, w \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$, we have

$$\sigma'(v + \lambda w) = \sigma' v + \lambda \sigma' w$$

For this, is is enough to show that the length $\ell(\sigma'(v + \lambda w) - \sigma' v - \lambda \sigma' w)$ is 0. The following calculation achieves this.

$$\begin{aligned} \ell(\sigma'(v + \lambda w) - \sigma' v - \lambda \sigma' w)^2 &= \ell(\sigma'(v + \lambda w) - \sigma' v)^2 + \ell(\lambda \sigma' w)^2 - 2\langle \sigma'(v + \lambda w) - \sigma' v, \lambda \sigma' w \rangle \\ &= d(\sigma'(v + \lambda w), \sigma' v)^2 + \lambda^2 \ell(\sigma' w)^2 - 2\langle \sigma'(v + \lambda w) - \sigma' v, \lambda \sigma' w \rangle \end{aligned}$$

Observe that $2\langle \sigma'(v+\lambda w) - \sigma' v, \lambda \sigma' w\rangle = 2\lambda\langle \sigma'(v+\lambda w), \sigma' w\rangle - 2\lambda\langle \sigma' v, \sigma' w\rangle$. Since dot product is invariant under σ', this is equal to $2\lambda\langle v+\lambda w, w\rangle - 2\lambda\langle v, w\rangle$. This, due to the bilinearity of dot product, is equal to $2\lambda\langle v+\lambda w - v, w\rangle = 2\lambda\langle \lambda w, w\rangle = 2\lambda^2 \ell(w)^2$. Therefore, we have

$$\begin{aligned}
\ell(\sigma'(v+\lambda w) - \sigma' v - \lambda\sigma' w)^2 &= d(\sigma'(v+\lambda w), \sigma' v)^2 + \lambda^2 \ell(\sigma' w)^2 - 2\lambda^2 \ell(w)^2 \\
&= d(v+\lambda w, v)^2 + \lambda^2 \ell(w)^2 - 2\lambda^2 \ell(w)^2 \\
&\quad \text{(since } \sigma \text{ preserves } d \text{ and } \ell\text{)} \\
&= \ell(v+\lambda w - v)^2 + \lambda^2 \ell(w)^2 - 2\lambda^2 \ell(w)^2 \\
&= \ell(\lambda w)^2 + \lambda^2 \ell(w)^2 - 2\lambda^2 \ell(w)^2 \\
&= \lambda^2 \ell(w)^2 + \lambda^2 \ell(w)^2 - 2\lambda^2 \ell(w)^2 = 0
\end{aligned}$$

This confirms the linearity of σ'

The situation is now reduced to show that the linear map $\sigma' : \mathbb{R}^2 \to \mathbb{R}^2$ comes from an orthogonal matrix. We will continue with this in the next lecture. $\square$

Lecture 5

Orthogonal matrices uphold bones of $\mathbb{R}^n$

January 15, 2025

Orthopaedic surgeons try to preserve relative distances between our bone cells. Orthogonal matrices also do the same! In fact, orthogonal matrices do much more by rotating and reflecting our bones!

Recall that an invertible matrix A is called an *orthogonal* matrix if $A^t = A^{-1}$. The set $\mathrm{O}_n(\mathbb{R})$ of $n \times n$ orthogonal matrices forms a group under matrix multiplication. Note that if $A \in \mathrm{O}_n(\mathbb{R})$, then $\det(A) \in \{-1, 1\}$.

5.1. Why should $A^t = A^{-1}$ govern distance preserving?

Though our bones are not planar, let us focus on matrices in $\mathbb{R}^2$. Arguments are the same for $\mathbb{R}^n$, when $n \geq 3$. Let $\tau : \mathbb{R}^2 \to \mathbb{R}^2$ be a distance preserving linear map. Since it is linear, $\tau(v) = Av$ for some 2×2 matrix A; and as we have seen in the last lecture, demanding τ to preserve distance can be rephrased as

$$\langle \tau v, \tau w \rangle = \langle v, w \rangle \quad \text{for all } v, w \in \mathbb{R}^2$$

Therefore, we have

$$v^t(A^tA)w = (Av)^tAw = \langle Av, Aw \rangle = \langle v, w \rangle = v^tw \quad \text{for all } v, w \in \mathbb{R}^2 \tag{5.1}$$

Let $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Let us denote the ij^{th} entry of A^tA by λ_{ij} so that

$$A^tA = \begin{pmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{pmatrix}$$

Then, putting $v = e_i$, $w = e_j$, and writing A^tA in terms of its entries in equation 5.1, we obtain $\lambda_{ij} = e_i^t(A^tA)e_j = e_i^te_j = \delta_{ij}$. Recall that the notation δ_{ij} stands for 1, if $i = j$; and 0, if $i \neq j$. This implies

$$A^tA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

and therefore $A^t = A^{-1}$.

In Theorem 4.1, the map $\sigma' := T_{-u} \circ \sigma$ was distance preserving. By the above discussion, σ' comes from an orthogonal matrix. Thus, the above discussion completes the proof of that theorem, and establishes that every symmetry of $\mathbb{R}^2$ is of the form $T_u \circ A$.

Why the name orthogonal is justified for such matrices? Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$. We expand the equation $A^t A = I_2$ to

$$\begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Let $v_1 = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$ and $v_2 = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$.

Then the above equation translates to

$$\begin{pmatrix} \langle v_1, v_1 \rangle & \langle v_1, v_2 \rangle \\ \langle v_2, v_1 \rangle & \langle v_2, v_2 \rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and we conclude that the columns of orthogonal matrices form an orthonormal set of $\mathbb{R}^2$, that is, both v_1 and v_2 have length 1 and the two vectors are orthogonal to each other. In fact, by considering the equation $AA^t = I_2$ we arrive at the conclusion that rows of orthogonal matrices also form an orthonormal set of $\mathbb{R}^2$.

5.2. Looking inside $\mathrm{O}_2(\mathbb{R})$

How does a 2×2 orthogonal matrix look like? $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, then the condition $A^t A = I_2$ poses the following restrictions

$$\begin{aligned} a_{11}^2 + a_{21}^2 &= 1 \\ a_{12}^2 + a_{22}^2 &= 1 \\ a_{11}a_{12} + a_{21}a_{22} &= 0 \end{aligned}$$

Thus, there exist $\alpha, \beta \in [0, 2\pi)$ such that

$$a_{11} = \cos\alpha, \quad a_{21} = \sin\alpha, \quad a_{12} = \sin\beta, \quad a_{22} = \cos\beta$$

Substituting these in the third equation above,

$$\cos\alpha \sin\beta + \sin\alpha \cos\beta = 0,$$

that is, $\sin(\alpha + \beta) = 0$. In other words, $\alpha + \beta \in \{0, \pi\}$, and we have two cases.

Case 1. $\beta = -\alpha$. In this case we have $\sin\beta = -\sin\alpha$ and $\cos\beta = \cos\alpha$. Thus, in this case

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\beta \\ \sin\alpha & \cos\beta \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$$

Case 2. $\beta = \pi - \alpha$. In this case we have $\sin\beta = \sin\alpha$ and $\cos\beta = -\cos\alpha$. In this case

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\beta \\ \sin\alpha & \cos\beta \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & -\cos\alpha \end{pmatrix}$$

We denote $R_\alpha := \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$ and $F_\alpha := \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$.

Recall that the transformation given by $v \mapsto R_\alpha v$ is the rotation by angle α in anticlockwise direction, and the one given by $v \mapsto F_\alpha v$ is the reflection about a mirror kept along the line $y = \tan\alpha x$. Thus, R_α is called a *rotation matrix* and F_α is called a *reflection matrix*. What we have shown here is that if $A \in \mathrm{O}_2(\mathbb{R})$ then A is either a rotation matrix or a reflection matrix.

Observe that F_0 is the reflection about x-axis. Note that $F_{\alpha/2} = R_\alpha F_0$. We summarize this discussion to the following.

Proposition 5.1. *Let $A \in \mathrm{O}_2(\mathbb{R})$. Then there exists a unique $\alpha \in [0, 2\pi)$ such that either $A = R_\alpha$ or $A = R_\alpha F_0$.*

5.3. Revisiting Theorem 4.1

In what follows, we state various composition rules. These are left as an exercise for the reader.

Exercise 5.2 By a direct calculation, obtain the following composition rules for symmetries.

(i). $T_u T_v = T_{u+v}$.

(ii). $R_\alpha R_\beta = R_{\alpha+\beta}$.

(iii). $F_0^2 = 1$.

(iv). $R_\alpha T_u = T_{u'} R_\alpha$, where $u' = R_\alpha(u)$.

(v). $F_0 T_u = T_{u'} F_0$, where $u' = F_0(u)$.

(vi). $F_0 R_\alpha = R_{-\alpha} F_0$.

In light of Proposition 5.1 and Exercise 5.2, we revisit Theorem 4.1 to state a much precise version of it.

Theorem 5.3. *Let $\sigma : \mathbb{R}^2 \to \mathbb{R}^2$ be a symmetry of $\mathbb{R}^2$. Then, there exists a unique $u \in \mathbb{R}^2$, and a unique $\alpha \in [0, 2\pi)$ such that either $\sigma = T_u R_\alpha$ or $\sigma = T_u R_\alpha F_0$.*

Proof. It follows from Theorem 4.1 and Proposition 5.1 that $\sigma = T_u R_\alpha F_0^i$, for some $u \in \mathbb{R}^2$, $\alpha \in [0, 2\pi)$ and $i \in \{0, 1\}$. To establish uniqueness of this decomposition, assume that $\sigma = T_u R_\alpha F_0^i = T_v R_\beta F_0^j$ be two such decompositions. Then, we use Exercise 5.2 (i) to get

$$T_{u-v} R_\alpha F_0^i = T_{-v} T_u R_\alpha F_0^i = T_{-v} T_v R_\beta F_0^j = R_\beta F_0^j$$

Where does it take $0 \in \mathbb{R}^2$? To $u - v$, if we look at the left hand side, and to 0, if we look at the right hand side. Hence $u = v$. This establishes uniqueness of the translation vector u. Further, it yields

$$R_\alpha F_0^i = R_\beta F_0^j$$

Now, we compare determinant of these matrices to conclude that $i = j$. Having done that, $R_\alpha = R_\beta$ is immediate. Since, we restrict the range of α within the interval $[0, 2\pi)$, the uniqueness of α is also evident. $\square$

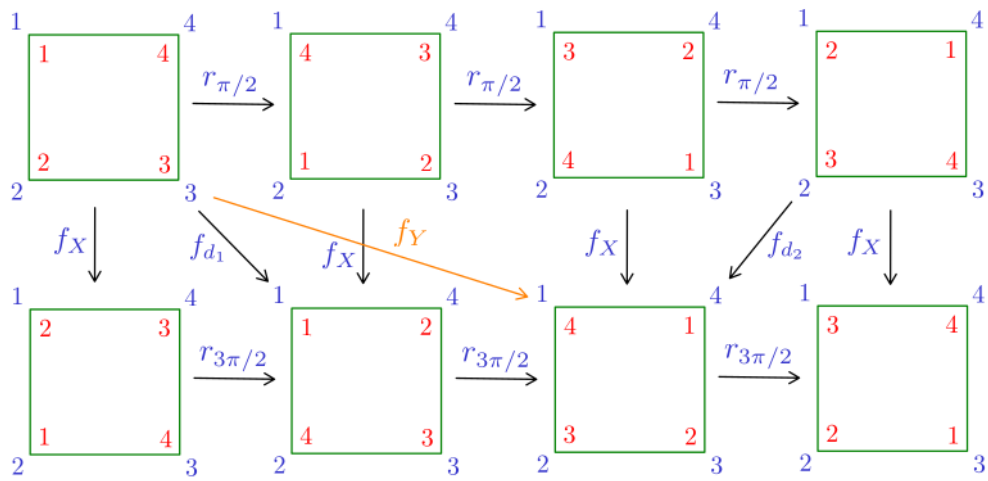
This allows us to make the following definition. A symmetry σ is called *orientation preserving* if σ is of the form $T_u R_\alpha$, and *orientation reversing*, if σ is of the form $T_u R_\alpha F_0$. That it is well defined is a consequence of Theorem 5.3.

Lecture 6

Symmetries of regular polygons: D_n

January 20, 2025

We have already seen symmetries of equilateral triangles and their composition table. The following depicts the symmetries of a square.



The composition table for squares is as follows.

	1	$r_{\pi/2}$	r_π	$r_{3\pi/2}$	f_X	f_Y	f_{d_1}	f_{d_2}
1	1	$r_{\pi/2}$	r_π	$r_{3\pi/2}$	f_X	f_Y	f_{d_1}	f_{d_2}
$r_{\pi/2}$	$r_{\pi/2}$	r_π	$r_{3\pi/2}$	1	f_{d_1}	f_{d_2}	f_Y	f_X
r_π	r_π	$r_{3\pi/2}$	1	$r_{\pi/2}$	f_Y	f_X	f_{d_2}	f_{d_1}
$r_{3\pi/2}$	$r_{3\pi/2}$	1	$r_{\pi/2}$	r_π	f_{d_2}	f_{d_1}	f_X	f_Y
f_X	f_X	f_{d_2}	f_Y	f_{d_1}	1	r_π	$r_{3\pi/2}$	$r_{\pi/2}$
f_Y	f_Y	f_{d_1}	f_X	f_{d_2}	r_π	1	$r_{\pi/2}$	$r_{3\pi/2}$
f_{d_1}	f_{d_1}	f_X	f_{d_2}	f_Y	$r_{\pi/2}$	$r_{3\pi/2}$	1	r_π
f_{d_2}	f_{d_2}	f_Y	f_{d_1}	f_X	$r_{3\pi/2}$	$r_{\pi/2}$	r_π	1

What is common between the composition tables of an equilateral triangle and a square?

6.1. Symmetries of regular polygons

In this lecture, we study symmetries of regular polygons. We first look at the following proposition.

Proposition 6.1. *Let R be a finite group of rotations in $\mathrm{O}_2(\mathbb{R})$. Then, there exists a rotation $r \in R$ and $m \in \mathbb{N}$ such that $R = \{1, r, r^2, \cdots, r^m\}$.*

Proof. Since each element in R is a rotation, we write

$$R = \{r_{\alpha_0}, r_{\alpha_1}, r_{\alpha_2}, \cdots, r_{\alpha_m}\}$$

Here $0 < \alpha_1 < \alpha_2 < \cdots < \alpha_m < 2\pi$, and $\alpha_0 = 0$. Therefore, r_{α_0} corresponds to the trivial rotation. The smallest denomination of the angle of a nontrivial rotation is α_1. Since R is a group, each $r_{\alpha_1}^i \in R$. We note that $r_{\alpha_1}^i = r_{i\alpha_1}$, and that $\alpha_i \leq i\alpha_1$.

We claim that $\alpha_i = i\alpha_1$ for every $i \in \{1, 2, \cdots, m\}$. To justify our claim, imagine the smallest i where $\alpha_i \neq i\alpha_1$. Denote such i by k. Then,

$$(k-1)\alpha_1 = \alpha_{k-1} < \alpha_k < k\alpha_1$$

Thus, $\alpha_k - \alpha_{k-1} < \alpha_1$. Observe that $r_{\alpha_k - \alpha_{k-1}} = r_{\alpha_k} r_{\alpha_{k-1}}^{-1} \in R$. Reading it along with $\alpha_k - \alpha_{k-1} < \alpha_1$ suggests that R contains a rotation of a denomination smaller than α_1. That is a contradiction, from which we achieve our claim $\alpha_i = i\alpha_1$. We follow a similar argument to obtain,

$$\alpha_m + \alpha_1 = m\alpha_1 + \alpha_1 = 2\pi.$$

Thus, $\alpha_1 = 2\pi/(m+1)$. To sum it up, we have $R = \{1, r, r^2, \cdots, r^m\}$ where $r = r_{2\pi/(m+1)}$. □

The group R is *generated* by one element, namely $r \in R$. Groups generated by single element are called *cyclic groups*. We will learn more about cyclic groups later. For now, we focus on symmetries of regular polygons.

Let us denote a regular polygon by P_n. We place the polygon within $\mathbb{R}^2$ in such a way that reflection about x-axis is a symmetry of P_n.

Theorem 6.2. *The group of symmetries of a regular polygon P_n is*

$$\mathrm{sym}(P_n) = \{1, r, r^2, \ldots, r^{n-1}, f, rf, r^2 f, \ldots, r^{n-1} f\},$$

where $r := r_{2\pi/n}$ denotes the rotation by angle $2\pi/n$ in anti-clockwise direction and $f := f_0$ denotes the reflection about x-axis. The composition table of $\mathrm{sym}(P_n)$ follows the following laws:

$$r^n = 1, \quad f^2 = 1, \quad fr = r^{-1} f$$

Proof. Let us justify it. First, by definition

$$\mathrm{sym}(P_n) := \{\sigma \in \mathrm{sym}(\mathbb{R}^2) : \sigma(S) = S\}$$

Thus, $\mathrm{sym}(P_n) \subseteq \mathrm{sym}(\mathbb{R}^2)$. Since P_n is a bounded shape, its group of symmetries cannot contain a nontrivial translation. Hence an element of $\mathrm{sym}(P_n)$ is either a reflection or a rotation. In other words, $\mathrm{sym}(P_n) \subseteq \mathrm{O}_2(\mathbb{R})$.

Let us focus on rotations first, and denote the set of rotations in $\mathrm{sym}(P_n)$ by $\mathrm{rot}(P_n)$. Note that the set $\mathrm{rot}(P_n)$ is itself a group. Since a rotation must map a vertex of P_n to a vertex of P_n, there are finitely many elements in $\mathrm{rot}(P_n)$. By Proposition 6.1, there is a rotation $r \in \mathrm{rot}(P_n)$ and $m \in \mathbb{N}$ such that $\mathrm{rot}(P_n) = \{1, r, r^2, \cdots, r^m\}$ where $r = r_{2\pi/(m+1)}$.

What is m in terms of n? Fix a vertex v_0 of P_n. Since $r \in \mathrm{rot}(P_n)$ is solely determined by where it takes v_0, the points $v_0, r(v_0), r^2(v_0), \cdots, r^m(v_0)$ are n distinct vertices of P_n. Thus, $m+1 = n$ and we have $\mathrm{rot}(P_n) = \{1, r, r^2, \cdots, r^{n-1}\}$. This completes our counting of rotations.

Now, about reflections: the polygon P_n is placed in such a way that f, the reflection about x-axis is a symmetry of P_n. That is, $f \in \mathrm{sym}(P_n)$. Since $\mathrm{sym}(P_n)$ is a group, we have $r^i f \in \mathrm{sym}(P_n)$ for every $i \in \{0, 1, \ldots, n-1\}$. These are n distinct reflections in $\mathrm{sym}(P_n)$. Have we missed accounting for some other reflection? For a moment, assume that we have indeed missed a reflection, say f'. Then, since composition of two reflections is a rotation, we have that $f'f$ is a rotation. Since our counting of rotations was complete, we have $f'f = r^i$ for some $i \in \{0, 1, \ldots, n-1\}$. Thus, $f' = f'ff = r^i f$, that we have already counted. This suggests that every reflection of P_n is of the form is $r^i f$, and that

$$\mathrm{sym}(P_n) = \{1, r, r^2, \ldots, r^{n-1}, f, rf, r^2 f, \ldots, r^{n-1} f\}$$

It is evident that $r^n = 1$, $f^2 = 1$. That $fr = r^{-1}f$ is precisely the content of Exercise 5.2 (vi). This completes the proof. □

Theorem 6.2 captures the composition tables of both, an equilateral triangle and a square. The group $\mathrm{sym}(P_n)$ is called a *dihedral group* of size $2n$.

6.2. Infinite dihedral group: $\mathrm{sym}(P_\infty)$

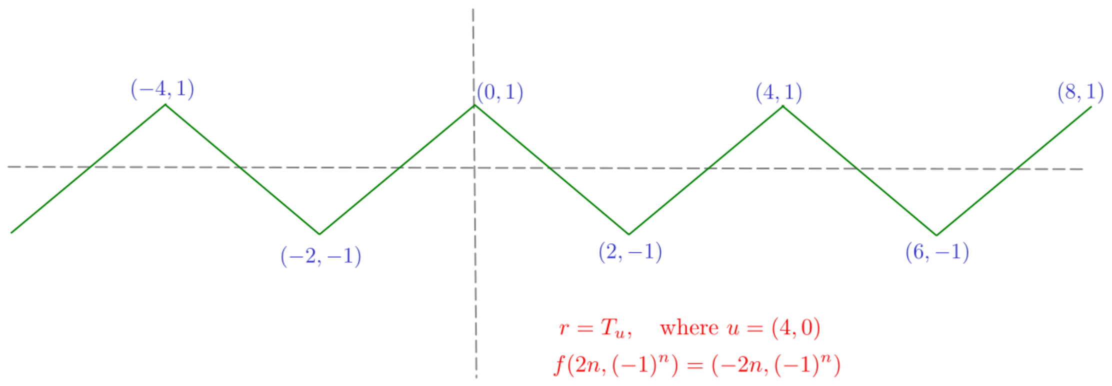
There is no polygon called P_∞. Nevertheless, let us think of an infinite version of dihedral groups. This will be obtained by considering rotations r for which $r^n \neq 1$ for every $n \in \mathbb{N}$. Are there such rotation? What about $r = r_\alpha$, where $\alpha = 2\pi/\sqrt{2}$?

We define the group $\mathrm{sym}(P_\infty)$ as follows:

$$\mathrm{sym}(P_\infty) := \{1, r^{\pm 1}, r^{\pm 2}, \ldots, f, r^{\pm 1} f, r^{\pm 2} f, \ldots, \}$$

where the composition table is determined by the relations $f^2 = 1$ and $fr = r^{-1}f$.

Is there some shape whose symmetry group is $\mathrm{sym}(P_\infty)$? What about the following zigzag shape?



$(-4,1)$
$(0,1)$
$(4,1)$
$(8,1)$
$(-2,-1)$
$(2,-1)$
$(6,-1)$
$r = T_u, \quad \text{where } u = (4,0)$
$f(2n,(-1)^n) = (-2n,(-1)^n)$

Lecture 7

Generating sets and Cayley graphs

January 22, 2025

In Lecture 6, we saw that the group $\mathrm{sym}(P_n)$ of symmetries of a regular n-gon is completely determined by two symmetries: a rotation r with $r^n = 1$, a reflection f (with, of course, $f^2 = 1$), and $fr = r^{-1}f$. In such a set up, we say that the subset $S := \{r, f\} \subseteq \mathrm{sym}(P_n)$ is a generating set of $\mathrm{sym}(P_n)$, and $r^n = 1, f^2 = 1, fr = r^{-1}f$ are the relations between elements of this generating set. In this lecture, we expose ourselves to the idea of generators and relations in a group and use it to give an interesting visualization of composition laws through *Cayley graphs*.

7.1. Generators and relations

Consider a group G and a subset $S \subseteq G$. Is S itself a group under the rule of composition that it inherits from G? Well, it depends on what S is. For example, if $G := \mathrm{sym}(P_n)$ and $S := \mathrm{rot}(P_n)$, then S is a group since the composition of two rotations is a rotation, and so is the inversion of a rotation. However, $S := \{r, rf, r^2\}$ is not a group, since it is devoid of the inverse of r, the composition $(rf)r$ (which is equal to f), and above all $1 \notin S$.

Let us take another example. Let $G = (\mathbb{R}, +)$ and $S \subseteq \mathbb{R}$ be the subset of irrational numbers. Is S a group? No, and you can list numerous reasons to justify this. But, if S is the subset of rational numbers, then it is a group under addition.

If a subset $S \subseteq G$ is itself a group under the rule of composition that it inherits from G, then S is called a *subgroup* of G. The set of even numbers is a subgroup of $(\mathbb{Z}, +)$.

Consider a subgroup H containing S, that is, $S \subseteq H \subseteq G$, where H is a subgroup of G. We denote $H \leq G$ to denote that H is a subgroup of G. The *smallest* subgroup H such that $S \subseteq H \leq G$ is called the subgroup *generated* by S. The subgroup generated by S is denoted by $\langle S \rangle$. Thus, if $G = \mathrm{sym}(P_n)$ and $S = \{f\}$, then $\langle S \rangle = \{1, f\}$.

Exercise 7.1

1. Let α be an irrational number. Let $r = r_{2\pi\alpha}$ denote the rotation by angle $2\pi\alpha$ in anti-clockwise direction and $f = f_0$ be the flipping about x-axis. Let $S := \{r, f\}$. Then, show that the subgroup $\langle S \rangle$ of $\mathrm{O}_2(\mathbb{R})$ is a proper subset of $\mathrm{O}_2(\mathbb{R})$.

2. Let S be a subgroup of $(\mathbb{Z}, +)$. Show that $S = \{ni : i \in \mathbb{Z}\}$ for some $n \in \mathbb{N}$.

A subset $S \subseteq G$ is called a *generating set* of G if $\langle S \rangle = G$. Note that $\{f_X, f_{d_1}\}$ is a generating set of $\mathrm{sym}(P_4)$ but $\{f_X, f_Y\}$ is not. What is missing in $\langle\{f_X, f_Y\}\rangle$? Check for f_{d_1}.

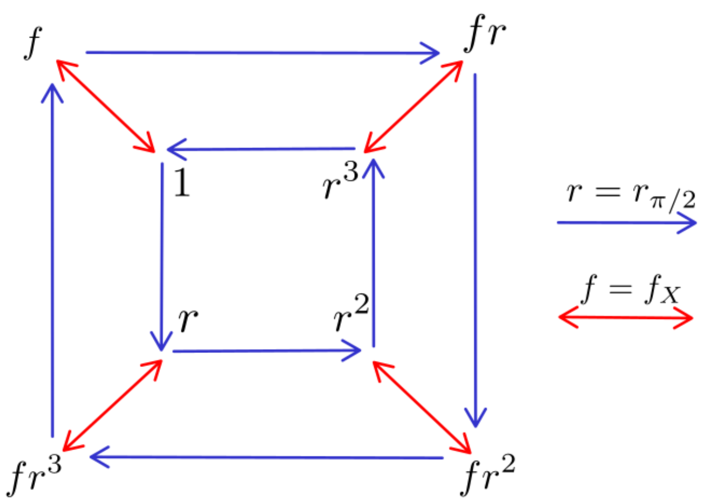
7.2. Cayley graphs

We can express interactions between group elements not just through composition tables or relations, but interesting graphs too. Let G be a group and S be a generating set of G. The *Cayley graph* $\Gamma(G, S)$ is the graph with the following construction.

1. Nodes of $\Gamma(G, S)$ are labeled with elements of G.
2. For $s \in S$, a colour c_s is assigned in a bijective manner.
3. For every $g \in G$ and $s \in S$, there is a directed edge of color c_s originating at node labeled g and terminating at the node labeled gs.

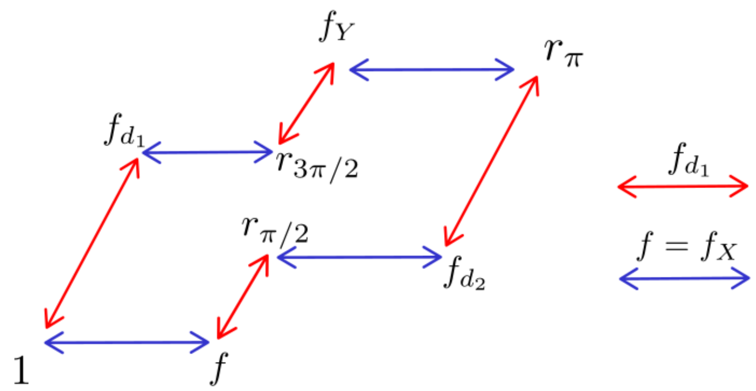
Exercise 7.2 Show that a Cayley graph $\Gamma(G, S)$ is a connected graph. In fact, given any $g \in G$, there exists a sequence of directed edges in $\Gamma(G, S)$ that starts at the node labeled 1 and terminates at the node labeled g.

Let us see some examples of Cayley graphs.

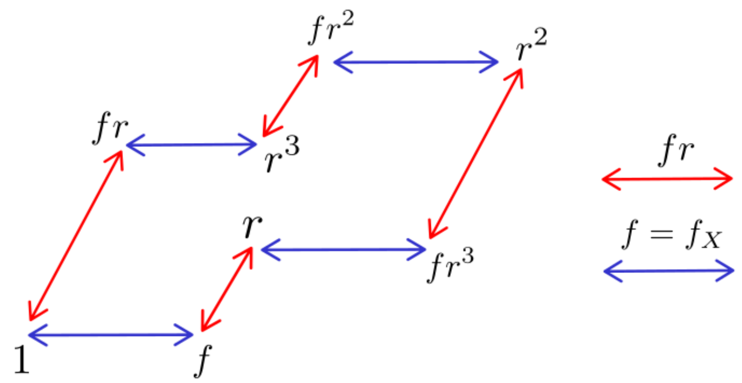


Cayley graph of $\mathrm{sym}(P_n)$ with $S = \{f, r\}$

The graph $\Gamma(G, S)$ not only depends on G, but also on the choice of generating set S. The following graph demonstrates this point.

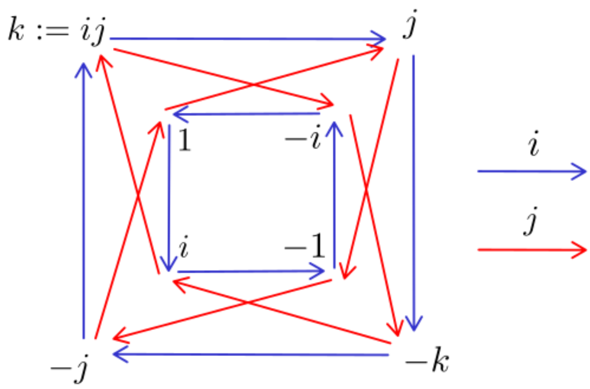


Cayley graph of $\mathrm{sym}(P_n)$ with $S = \{f, f_{d1}\}$



Relabeling above Cayley graph in terms of f and r. Here $S = \{f, fr\}$.

Finally, it is possible to write down the composition table from $\Gamma(G, S)$. Try your hands on the following Cayley graph.



Cayley graph of quaternion group Q_2 with $S = \{i, j\}$

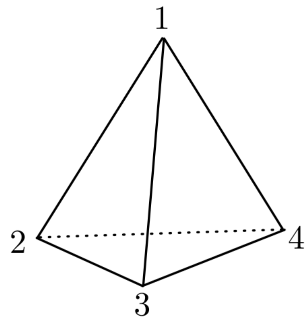
You will see the composition table that you derive from this graph has a flavour of the cross product of the three vectors $\hat{i}, \hat{j}, \hat{k}$ that you would have encountered in your class XII Physics.

Lecture 8

Symmetries of a Tetrahedron

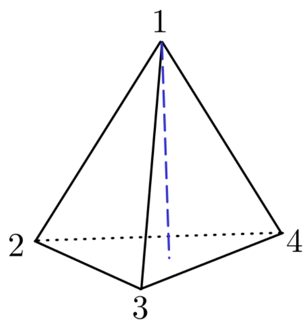
January 27, 2025

Three equidistant points in $\mathbb{R}^2$, when connected through edges, form an equilateral triangle. Four equidistant points in $\mathbb{R}^3$, when connected through edges, form a regular tetrahedron. We usually drop the adjective *regular* and call it a simply a tetrahedron–four faced solid. In this lecture, we wish to study symmetries of a tetrahedron. Let us denote a tetrahedron by $\mathbb{T}$.



A tetrahedron with vertices labeled

Let us think of simplest nontrivial symmetries–rotations that keep the top vertex fixed. Axis of such rotations passes through the top vertex and the centroid of the triangle $\Delta(2 \to 3 \to 4)$ opposite to it. There are two such rotations. At this point we need to fix a convention. Our rotations will appear anti-clockwise as seen from the vertex 1 toward the opposite triangle $\Delta(2 \to 3 \to 4)$. The angles of these rotations are $2\pi/3$ and $4\pi/3$. We denote the one by angle $2\pi/3$ by p. Then the other rotation will be p^2. Of course, p^3 is the trivial rotation, but here we are focusing on nontrivial rotations. The axis of the rotations p and p^2 is marked below.



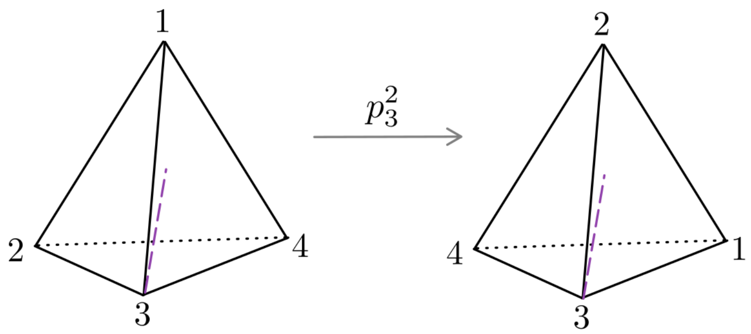
The axis of the rotation of p and p^2

Now, let us vary the vertex of rotation. We have four vertices available to carry out such rotations. Let us denote by p_i the rotation by angle $2\pi/3$ along the axis passing through vertex labeled i. As per the convention that we have fixed, the sense of this rotation is anti-clockwise when seen from the vertex i toward the opposite face. In this notation p that we described above in detail is to be denoted p_1.

Thus far, how many rotations have been accounted for? Let us list them here:

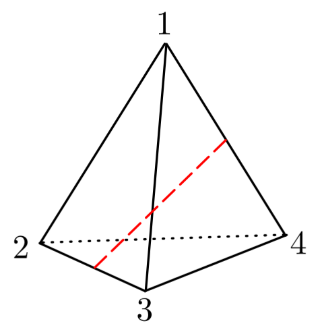
$$p_1, p_1^2, p_2, p_2^2, p_3, p_3^2, p_4, p_4^2$$

To be convinced that you have followed this notation, note that the following rotation is p_3^2. Here, the axis passes through the vertex 3 and the centroid of the opposite face $\Delta(1 \to 2 \to 4)$.



The rotation p_3^2

Now, let us identify one more type of rotation. The axis of this rotation is the mid point of opposite edges.



The axis for a rotation of order 2

A rotation by angle π along the axis as shown in this picture is also a symmetry. Let us denote it by ℓ. This symmetry swaps vertex 1 with vertex 4, and the vertex 2 with vertex 3. We denote $\ell : (1 \leftrightarrow 4)(2 \leftrightarrow 3)$ to express this impact of ℓ on the tetrahedron.

There are three such rotations. Let us list them here, along with their impact on the tetrahedron.

$$\begin{aligned}\ell &: (1 \leftrightarrow 4)(2 \leftrightarrow 3)\\ \ell' &: (1 \leftrightarrow 3)(2 \leftrightarrow 4)\\ \ell'' &: (1 \leftrightarrow 2)(3 \leftrightarrow 4)\end{aligned}$$

And finally, lets not forget the trivial symmetry. Putting these rotational symmetries a set:

$$\mathcal{A} := \{1, p_1, p_1^2, p_2, p_2^2, p_3, p_3^2, p_4, p_4^2, \ell, \ell', \ell''\}$$

What we have shown so far is that $\mathcal{A}$ is a subset of $\mathrm{rot}(\mathbb{T})$. Here, $\mathrm{rot}(\mathbb{T})$ denotes the set of rotational symmetries of a tetrahedron.

Question 8.1 Is the set $\mathrm{rot}(\mathbb{T})$ actually a group? That is, if we compose two rotational symmetries with *different* axes of rotation, is the resultant also a rotation?

Answer to this questions is, yes! We are postponing an explanation to this, but you are encouraged to think about it. It is interesting to think of an interplay of axes while composing rotational symmetries about different axes.

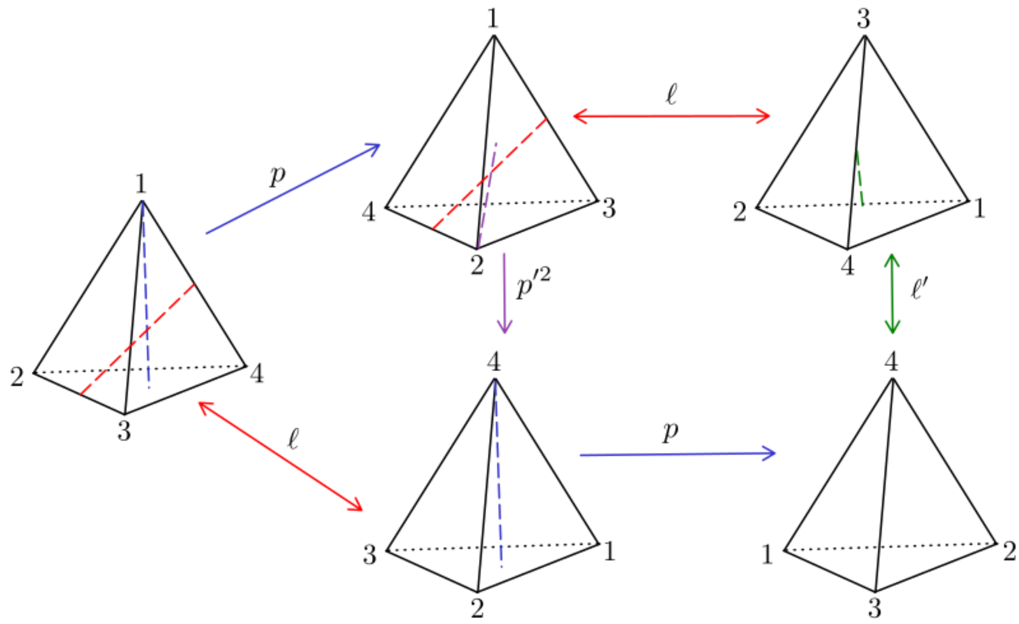
Question 8.2 Is $\mathcal{A}$ a *subgroup* of $\mathrm{rot}(\mathbb{T})$?

It is clear that $\mathcal{A}$ is closed under inversion, order 2 symmetries are self-inverses, and for each $i \in \{1, 2, 3, 4\}$, the symmetries p_i and p_i^2 are inverses of each other.

Alert

While composing symmetries, we have to ensure that the frame of reference is drawn from the *identity position* of the tetrahedron.

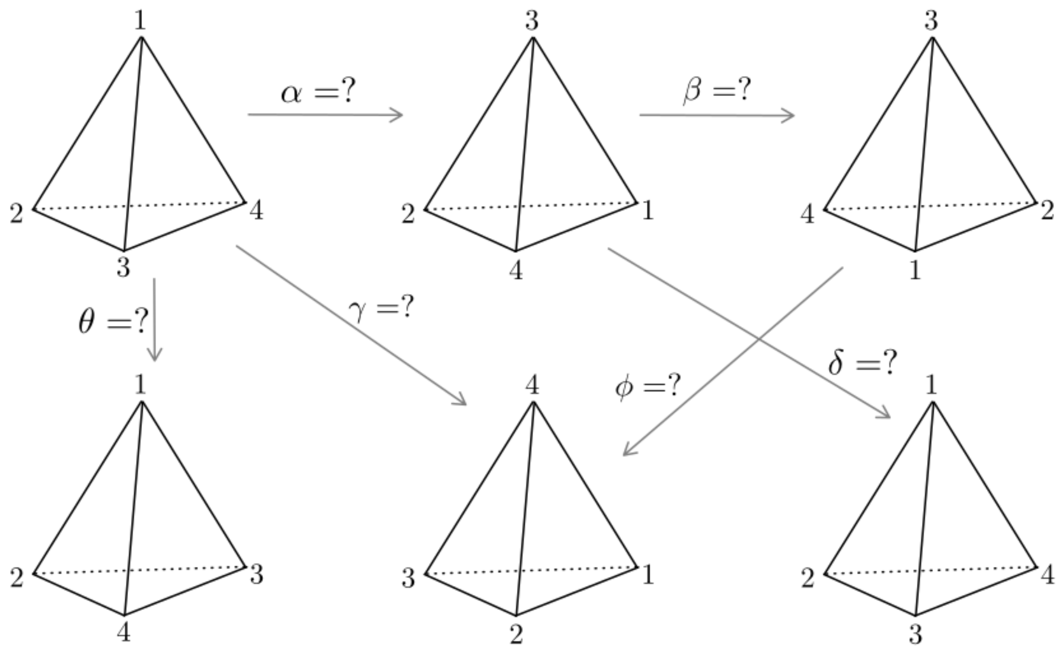
Through the following picture we explore various combinations of compositions of elements of $\mathcal{A}$.



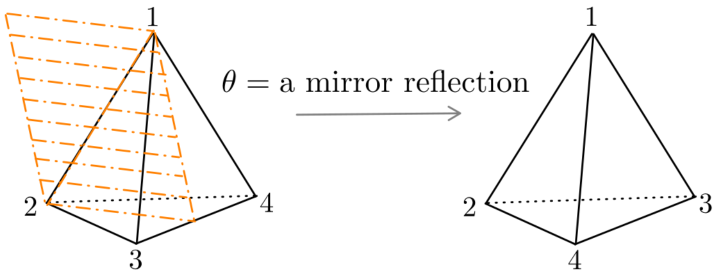
Composition of some symmetries in $\mathcal{A}$

Exercise 8.3 Write the composition table of $\mathcal{A}$.

Exercise 8.4 In the following picture identify which symmetries are $\alpha, \beta, \gamma, \delta, \phi$ and θ.



Which of the symmetries in $\mathcal{A}$ is θ? None! Is it even a symmetry? Look at the following picture to convince yourself that θ is reflection about a mirror kept on the plane passing thorugh the edge $1 \leftrightarrow 2$ the mid point of the edge $3 \leftrightarrow 4$. Though θ is a symmetry, it does not belong to $\mathcal{A}$. It is not a rotational symmetry.



How many such symmetries are there? In other words, what are all possible placements of such mirrors? Any such mirror passes through an edge and the mid point of the opposite edge. There are 6 edges, and therefore, there are 6 such mirror reflections.

This make the count $12 + 6 = 18$. Of these, 12 symmetries are rotational and 6 are reflections. Any other symmetry?

Shall we use elements of $\mathcal{A}$ and θ to generate some symmetries? Here it is.

$$\theta A := \{\theta, \theta p_1, \theta p_1^2, \theta p_2, \theta p_2^2, \theta p_3, \theta p_3^2, \theta p_4, \theta p_4^2, \theta \ell, \theta \ell', \theta \ell''\}$$

That's 12 more symmetries in just one go! So, should we say that we have counted $12 + 6 + 12 = 30$ symmetries so far? There is a possibility of double counting. And indeed, we have counted some symmetries twice!

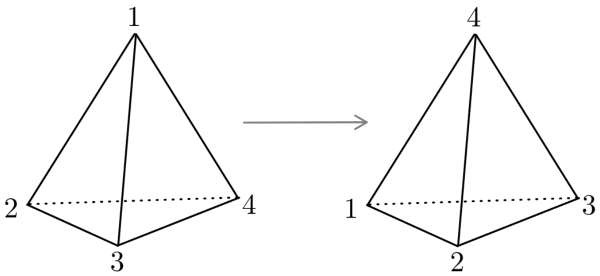
Any symmetry of a tetrahedron permutes four vertices. Thus the number of symmetries cannot exceed $24 = 4 \times 3 \times 2 \times 1$. That means, we have fallen into the trap of double counting.

Exercise 8.5 Show that the set $\theta\mathcal{A}$ contains 12 elements and $\mathcal{A}$ is disjoint from $\theta\mathcal{A}$.

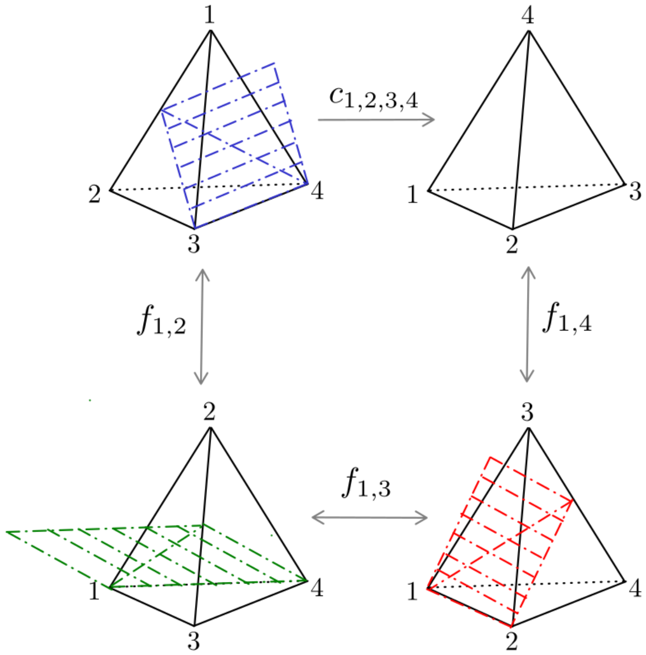
This exercise shows that $\mathcal{A} \cup \theta\mathcal{A}$ captures all 24 permutations of 4 vertices as symmetries of a tetrahedron.

What about the 6 mirror symmetries that we found, intelligently? These must be hidden somewhere in the set $\theta\mathcal{A}$.

Finally, there must be 6 symmetries which are neither a reflection, nor a rotation! Lets look at this.



Does it have an axis of rotation? Does it have a mirror of reflection? No. It is one of the symmetries which is neither a rotation nor a reflection. However, it is a composition of three reflections. The following picture demonstrates this point.



Neither a reflection, nor a rotation : $c_{1,2,3,4} = f_{1,4}f_{1,3}f_{1,2}$

Lecture 9

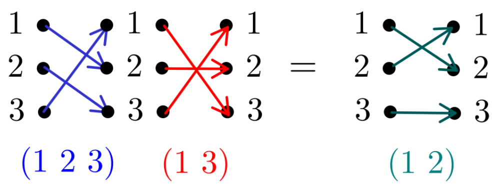
Permutations

January 29, 2025

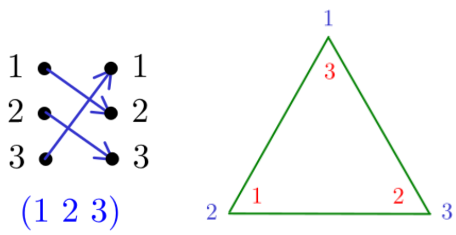
In all the lectures so far, we would have noticed that a symmetry *permutes* some 'crucial ingredients' of the shape in question. These ingredients are vertices, edges, faces – or even something that is not so apparent! Therefore, one may hope to understand symmetry via permutations. This lecture takes us a step closer to this hope.

9.1. What is a permutation

Let $n \in \mathbb{N}$, and $N := \{1, 2, 3, \cdots, n\}$ be the set of first n natural numbers. A bijection $\sigma : N \to N$ is called a *permutation* on N. Recall that a bijection is a one-one onto map. Let S_n denote the set of all permutations on N. Note that S_n has $n! = n \times (n-1) \times \cdots \times 2 \times 1$ elements. Two permutations can be composed together to obtain a third one, and a permutation can be '*unpermuted*' to restore the initial one. Hence S_n is a group. Following is an example of composition of permutations.



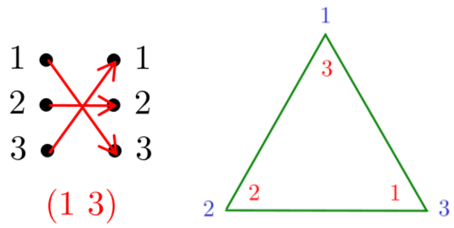
Let us understand what is happening here. First, the permutation $\tau := (1\ 2\ 3)$ is the bijection $\tau(1) = 2$, $\tau(2) = 3$, $\tau(3) = 1$. Similarly, the permutation $\sigma := (1\ 3)$ is the bijection $\sigma(1) = 3$, $\sigma(2) = 2$, $\sigma(3) = 1$. We may depict these permutations through the symmetries of a triangle. The permutation τ corresponds to the following orientation of the triangle:



Rotation $r_{2\pi/3}$ has the same effect as $\tau = (1\ 2\ 3)$

The way to read it is the following: the vertex 1, under the permutation, occupies the original position of the vertex 2. Similarly, the vertex 2 occupies the original position of the vertex 3, and the vertex 3 occupies the original position of the vertex 1. Observe that this corresponds to the rotational symmetry $r_{2\pi/3}$.

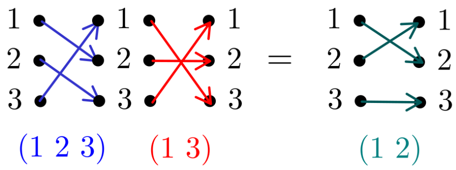
Along these lines, the permutation σ corresponds to the following triangle:



Reflection f_2 has the same effect as $\sigma = (1\ 3)$

This corresponds to the reflection symmetry f_2.

Now, let us go back to the composition of permutations.



The permutation $\tau = (1\ 2\ 3)$ takes place first, and is followed by the permutation $\tau = (1\ 3)$. The net outcome of the composition is the permutation $\tau = (1\ 2)$. Hence the composition is written as

$$(1\ 3)(1\ 2\ 3) = (1\ 2).$$

It is important to pay attention to the order in which we write it. The action that happens first is placed rightmost in the notation of the composition. What does $(1\ 3)(1\ 2\ 3) = (1\ 2)$ correspond to, in terms of symmetry of a triangle? It is $f_2 r_{2\pi/3} = f_3$. Go back to the composition table of symmetries of triangle and confirm that indeed $f_2 r_{2\pi/3} = f_3$. That means, composition of the permutation is same as the composition of symmetries. Thus, we can use the group S_n to understand symmetries in $\mathbb{R}^2$. This idea is not restricted to $\mathbb{R}^2$. Tetrahedron symmetries may be understood using the group S_4 by carefully tracking the movement of four vertices of the tetrahedron. In fact, any shape whose group of symmetries is finite can be understood in terms of permutations.

Before we try our hands on tetrahedron, let us learn more about permutations and their composition.

Definition 9.1. *Let $1 \leq r \leq n$. A permutation $\sigma \in S_n$ is called an r-*cycle *if there exists an ordered r-tuple $(a_1, a_2, \ldots, a_r)$, with each $a_i \in \{1, 2, \cdots, n\}$ distinct, such that $\sigma(m) = m$ for every $m \notin \{a_1, a_2, \ldots, a_r\}$, and*

$$\sigma(a_i) = \begin{cases} a_{i+1} & \text{if } 1 \leq i \leq r-1, \\ a_1 & \text{if } i = r. \end{cases}$$

We denote this r-cycle by $(a_1\ a_2\ \dots\ a_{r-1}\ a_r)$*.*

It is important to note that $(a_1\ a_2\ \dots\ a_{r-1}\ a_r)$ and $(a_2\ a_3\ \dots\ a_r\ a_1)$ represent the same r-cycle. A permutation is called a cycle if it is an r-cycle for some $r \le n$.

What are all 1-cycles? There is only one–the identity permutation. A 2-cycle swaps two symbols and does not affect the rest. The permutations in S_3 that correspond to reflections in triangles are 2-cycles. Observe that every $\sigma \in S_3$ is either a 1-cycle, a 2-cycle, or a 3-cycle. Is it true that every permutation in S_n is an r-cycle for a suitable r? What about S_4? Consider the permutation $\tau \in S_4$ that swaps 1 with 2, and 3 with 4? It is not a cycle. However τ is a *composition* of cycles $(1\ 2)$ and $(3\ 4)$. A permutation that is not a cycle is called a *non cycle*.

Exercise 9.2

1. Show that S_4 contains exactly 3 permutations which are not cycles.
2. Find all non cycles in S_5.
3. Is there $n \in \mathbb{N}$ such that S_n contains more non cycles than cycles? If yes, find minimal such n.
4. How does the number of cycles and non cycles grow with n?

Two cycles $(a_1\ a_2 \dots\ a_{r-1}\ a_r)$ and $(b_1\ b_2 \dots\ b_{s-1}\ b_s)$ are called *disjoint* if the subsets $\{a_1, a_2, \dots, a_{r-1}, a_r\}$ and $\{b_1, b_2, \dots, b_{s-1}, b_s\}$ of N are disjoint. For example, $(1\ 4)$ and $(2\ 3)$ are disjoint cycles, but $(1\ 2)$ and $(2\ 3)$ are not.

The following theorem shows the power of composition.

Theorem 9.3. *Every $\sigma \in S_n$ can be expressed as a composition of disjoint cycles.*

Proof. The proof is quite intuitive and constructive. We start with 1 and traverse through images of σ and its powers, until we come back to 1. Since N is a finite set, coming back to 1 after finitely many steps is inevitable. That is, $\sigma^r(1) = 1$ for a suitable $r \le n$. Here, σ^2 stands for composition σ with itself. Similarly, σ^r is the composition of σ^{r-1} and σ. So, our route while traversing through elements of N is

$$1 \to \sigma(1) \to \sigma^2(1) \to \cdots \to \sigma^{r-1}(1) \to \sigma^r(1) = 1$$

It will be fun to call this route the *orbit* of 1. Denote $a_i := \sigma^{i-1}(1)$ and observe that the r-cycle $(a_1\ a_2\ \dots\ a_r)$ captures the impact of σ on elements falling in this orbit. Do we conclude that $\sigma = (a_1\ a_2\ \dots\ a_r)$? No! There are other elements—the ones that have not yet fallen into the orbit, and on whom we have to assess the impact of σ. So, repeat! Find the smallest element, say, i, in N that is yet to be encountered in an orbit, and consider the orbit of this element. Note that this orbit is disjoint from the previous one. A suitable cycle will capture the impact of this orbit. We continue till all elements of N are exhausted, and compose all cycles that we find on the way. Voila! We have written σ as composition of disjoint of cycles. $\square$

What is even more intriguing is that every cycle can be written as a composition (though, not necessarily disjoint) of 2-cycles. The following composition exhibits this point.

$$(a_1\ a_2 \ldots\ a_{r-1}\ a_r) = (a_1\ a_r)(a_1\ a_{r-1}) \cdots (a_1\ a_3)(a_1\ a_2)$$

An alternate terminology for a 2-cycle is a *transposition*. We assimilate this discussion into the following theorem.

Theorem 9.4. *Every $\sigma \in S_n$ is a composition of transpositions.*

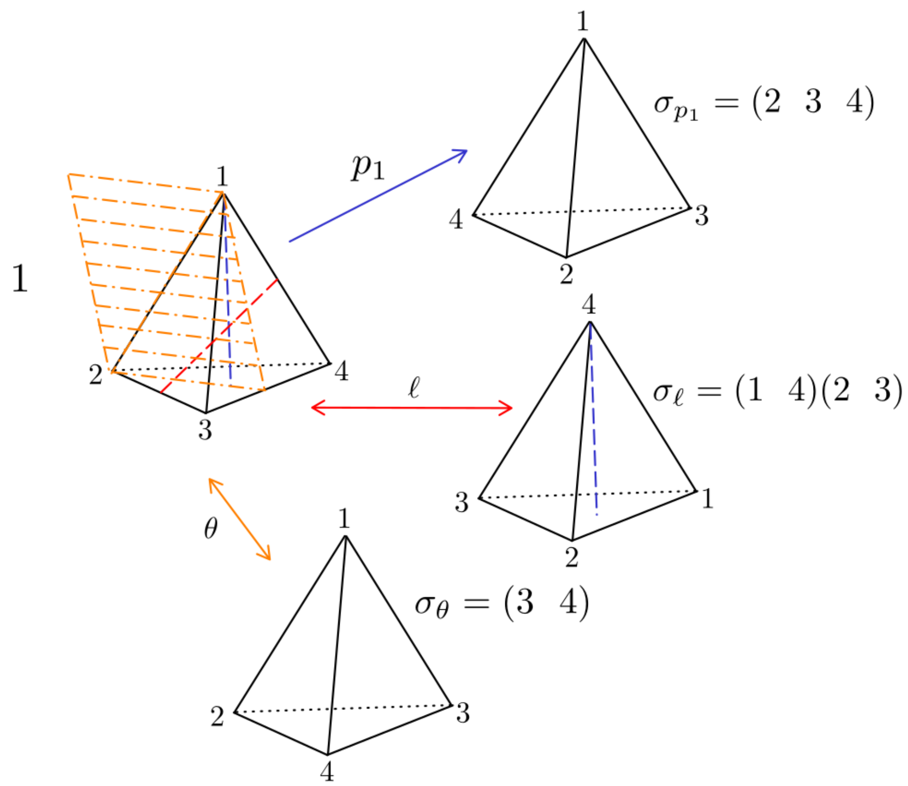
These transposition need not be disjoint.

It is indeed amazing to observe that to obtain any arbitrary permutation, one needs to pick two symbols at a time and swap them, and carefully repeat the process long enough. For example:

$$(1\ 2\ 3) = (1\ 3)(1\ 2)$$

In a sense, transpositions play the same role in understanding permutations as the one played by reflections in understanding symmetry. A careful study of symmetry of tetrahedron using permutations of vertices will clearly exhibit this philosophy.

9.2. Symmetry of tetrahedron and permutations



This picture depicts the symmetries p_1, ℓ and θ in terms of permutations. The way to write permutation σ_{p_1} is as follows: Since under the transformation p_1, the vertex 1 does not change its location, we do not write it in σ_{p_1}. The vertex 2, under the transformation p_1 occupies the original position of the vertex 3, so we write $(2\ \ 3$. Along the same lines, the vertex 3 occupies the position of the vertex 4, and we continue writing $(2\ 3\ 4$. The vertex 4 occupies the position of vertex 2 – so a cycle is complete, and we close the bracket to write $(2\ 3\ 4)$. There are no more vertices and thus the permutation corresponding to p_1 is $\sigma_{p_1} = (2\ 3\ 4)$.

In fact, given two arbitrary $\alpha, \beta \in \mathrm{sym}(\mathbb{T})$. Convince yourself that $\sigma_{\beta\alpha} = \sigma_\beta \sigma_\alpha$. This suggests that the symmetry group of a tetrahedron is *isomorphic* to S_4, and the isomorphism is given by

$$\begin{aligned} \mathrm{sym}(\mathbb{T}) &\to S_4 \\ \alpha &\mapsto \sigma_\alpha \end{aligned}$$

We are yet to formally encounter the concept of isomorphism between groups. Still, you may like to try the following exercise.

Exercise 9.5

1. Show that $\mathrm{sym}(\Delta)$, the symmetry group of an equilateral triangle, is isomorphic to S_3.
2. Show that $\mathrm{sym}(\square)$, the symmetry group of a square, is isomorphic to a subgroup of S_4.
3. Think of symmetries of a tetrahedron as the transformations that permute six edges. Enumerate those edges, follow how they permute under symmetries and find a subgroup of S_6 that is isomorphic to $\mathrm{sym}(\mathbb{T})$.

Lecture 10

More on Permutations

February 03, 2025

In the last lecture, we saw how to compose and decompose permutations. Particularly remarkable is the Theorem 9.4 which exhibits that every permutation can be obtained by successively swapping two symbols at a time.

In this lecture, we classify permutations into two types – *even* and *odd* permutations.

10.1. Signature of a permutation

Let $n \in \mathbb{N}$. Consider variables $x_1, x_2, \ldots, x_n$ and the following polynomial in these variables:

$$Q(x_1, x_2, \ldots, x_n) := \prod_{1 \le i < j \le n} (x_i - x_j)$$

In short, we denote this polynomial by Q. The polynomial Q is a product of $n(n-1)/2$ monomials. For $n = 3$, we have $Q = (x_1 - x_2)(x_1 - x_3)(x_2 - x_3)$. Let $\sigma \in S_n$. Define

$$\sigma Q := \prod_{1 \le i < j \le n} (x_{\sigma(i)} - x_{\sigma(j)})$$

The action of σ on indices may have resulted into many pairs (i, j) for which $i < j$, but $\sigma(i) > \sigma(j)$. Let $|\sigma|$ denote the number of such pairs. Then

$$\sigma Q := (-1)^{|\sigma|} \prod_{1 \le i < j \le n} (x_i - x_j)$$

Thus, given $\sigma \in S_n$ either $\sigma Q = Q$ or $\sigma Q = -Q$. We define a map $\operatorname{sgn} : S_n \to \{\pm 1\}$ as follows

$$\operatorname{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma Q = Q, \\ -1 & \text{if } \sigma Q = -Q. \end{cases}$$

A permutation $\sigma \in S_n$ is called an *even* permutation if $\operatorname{sgn}(\sigma) = 1$, and an *odd* permutation if $\operatorname{sgn}(\sigma) = -1$. The following exercise is a consequence of simple observations.

Exercise 10.1

1. If $\sigma, \tau \in S_n$ then $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$.

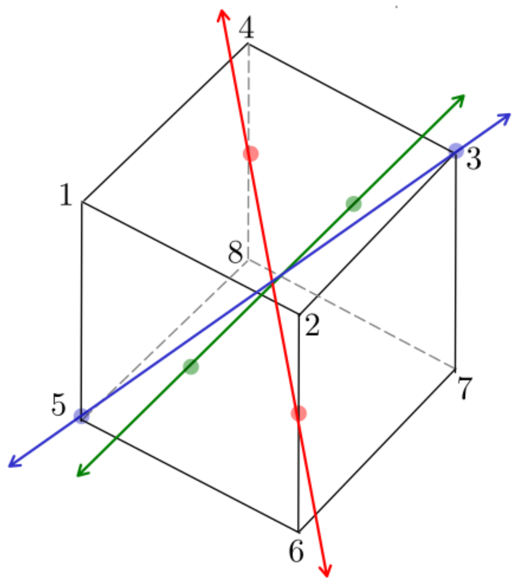
2. If σ is a transposition, then $\operatorname{sgn}(\sigma) = -1$.

3. If $\sigma = \tau_1\tau_2 \ldots \tau_m$, where each τ_i is a transposition then $\operatorname{sgn}(\sigma) = (-1)^m$.

4. If σ is an r-cycle then $\operatorname{sgn}(\sigma) = (-1)^{r-1}$.

10.2. Symmetries of a cube and permutations

We can explore permutations to understand symmetries of a cube – exactly the way we did for a tetrahedron. The three types of axes are shown in the following.



We have

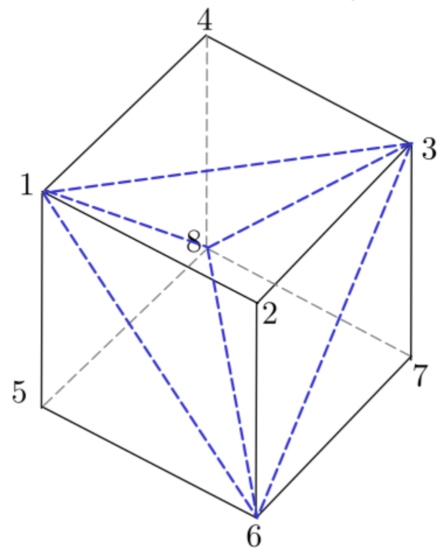
1. Three axes passing through the centres of opposite faces. Each such axis supports three nontrivial symmetries by angles $\pi/2$, π and $3\pi/2$. This accounts for $3 \times 3 = 9$ rotational symmetries of a cube.

2. Four axes passing through opposite vertices. Each such axis supports two nontrivial symmetries by angles $2\pi/3$ and $4\pi/3$. This accounts for $4 \times 2 = 8$ rotational symmetries of a cube.

3. Six axes passing through the mid points of opposite edges. Each such axis supports one nontrivial symmetry by the angle π. This accounts for $6 \times 1 = 6$ rotational symmetries of a cube.

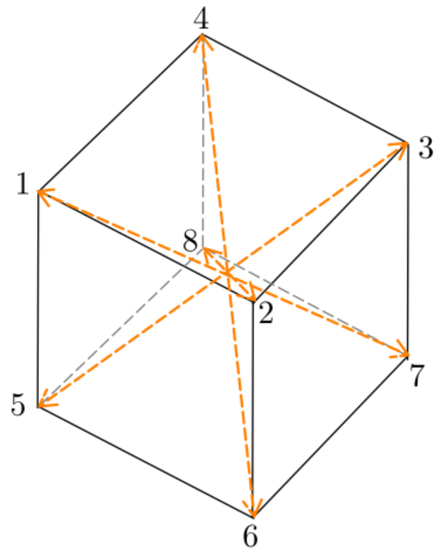
And, of course, there is identity symmetry. There are no other axes and we have accounted for all $9 + 8 + 6 + 1 = 24$ symmetries of a cube.

The following picture shows that we can inscribe a tetrahedron inside a cube.



Thus, each rotational symmetry of a tetraherdon can be thought of as a symmetry of a cube.

In the following picture, we show four diagonals of a cube.



A symmetry permutes these diagonals. Thus the group of rotational symmetries of a cube may be understood in terms of permutations of these four diagonals.

Lecture 11

Cube in a dodecahedron

February 10, 2025

We refer to slides for this lecture available at this link in the Google drive of the course. Another resource is this animation of a cube inscribed in a dodecahedron. Thanks to Daksh Arora ⟨ms23206@iisermohali.ac.in⟩ for creating this.

Lecture 12

Abstraction : A concrete foundation for organizing ideas

February 12, 2025

So far, we had shapes – the polygons, polyhedra, circles, spheres – and we studied their symmetry. We had the idea of compositing symmetries and the concept of groups came naturally. Subgroups captured the symmetries of some inscribed objects.

We now wish to turn the table. We will allow abstract ideas to take a lead and see how they lay a foundation for our further understanding of symmetry. It is assumed that the reader is familiar with the notions of a group, subgroup, a generating set of a group, and relations between elements in a generating set. You are encouraged to revise these.

12.1. Knowing groups from their subgroup

Let G be a group and H be a subgroup of G – we express it by writing $H \leq G$. We have encountered following instances of $H \leq G$ in this course.

1. $\mathrm{SO}_3(\mathbb{R}) \leq \mathrm{O}_3(\mathbb{R})$: 3×3 special orthogonal matrices form a subgroup of the group of 3×3 orthogonal matrices.
2. $A_n \leq S_n$: even permutations form a subgroup of the group of permutations.
3. $\mathrm{rot}(P_n) \leq \mathrm{sym}(P_n)$: rotational symmetries form a subgroup of the group of all symmetries of a regular n-gon.
4. $\mathrm{rot}(\mathcal{C}) \not\leq \mathrm{rot}(\mathcal{D})$: the group of rotational symmetries of a cube is **not** a subgroup of the group of rotational symmetries of a dodecahedron.
5. $(2\mathbb{Z}, +) \leq (\mathbb{Z}, +)$: even integers form a subgroup of additive group of all integers.

Given $H \leq G$, we construct special subsets of G call *cosets*. For $g \in G$ define

$$gH := \{gh : h \in H\}$$

Thus, upon composing all elements of H *on left* by g, we obtain gH. It is called the *coset of H in G represented by g*. In general, gH is not a subgroup of G. In the following proposition, we formulate some observations regarding cosets.

Proposition 12.1. *Let $H \leq G$. Then*

(i). gH and H are in bijection.

(ii). $gH = H$ if and only if $g \in H$.

(iii). $g_1H = g_2H$ if and only if $g_2^{-1}g_1 \in H$.

(iv). Let $g_1, g_2 \in G$. Then either $g_1H = g_2H$ or g_1H and g_2H are disjoint.

Proof. (i). The function $x \to g^{-1}x$ establishes a bijection from gH to H, and we are done.

(ii). Suppose, $gH = H$. Then there exist $h_1, h_2 \in H$ such that $gh_1 = h_2$. Therefore, $g = h_2h_1^{-1} \in H$. Conversely, if $g \in H$, then for every $h \in H$ we have $gh \in H$. Thus $gH \subseteq H$. Further, $g^{-1} \in H$, as H is a subgroup. A similar argument shows that $g^{-1}H \subseteq H$. Assimilating all this: $H = gg^{-1}H \subseteq gH \subseteq H$. Thus, $gH = H$.

(iii). We read part (ii) with $g = g_2^{-1}g_1$. Hence $g_2^{-1}g_1 \in H$ is equivalent to $g_2^{-1}g_1H = H$. This, in turn, is equivalent to $g_1H = g_2g_2^{-1}g_1H = g_2H$.

(iv). It is enough to assume that g_1H and g_2H are not disjoint and show that $g_1H = g_2H$. Suppose $z \in g_1H \cap g_2H$. Then $z = g_1x = g_2y$ form some $x, y \in H$. Consequently, $g_2^{-1}g_1 = yx^{-1} \in H$. Now, by part (iii), $g_1H = g_2H$.

□

We assimilate it into the following

Theorem 12.2. *Let G be a group and H be a subgroup of G. Then there exists a subset $A \subseteq G$ such that*

$$G = \bigsqcup_{g \in A} gH$$

Here $\bigsqcup$ denotes the disjoint union.

Proof. Let us read the statement aloud. *Given $H \leq G$, the group G is a disjoint union of cosets of H in G.* Since an arbitrary $g \in G$ belongs to a coset (the coset gH, obviously) it follows that G is a union of cosets of H in G. Now, from Proposition 12.1 (iii), it follows that the contribution of various cosets in this union is redundant. We remove this redundancy and obtain a disjoint union using Proposition 12.1 (iv). This completes the proof.

Alternatively, We could have approached it as follows. If $H = G$, then the statement is trivial. Otherwise we pick $g \in G \backslash H$ and consider gH. By Proposition 12.1 (iii), the cosets H and gH are disjoint and $H \sqcup gH$ is a disjoint union. We consider a maximal disjoint union of cosets of H in G. This union is equal to G, otherwise we can pick $g \in G$ which is not covered in this union. The union of the coset gH and this union contradicts maximality of the union. This suggests that the maximal disjoint union of cosets is equal to the whole group G. □

Let us work out an example. Take $G = S_3$ and $H = \{1, (1\,2)\}$. Let $g_1 = (1\,3)$, $g_2 = (1\,2\,3)$, and $g_3 = (1\,3\,2)$. Then,

$$\begin{aligned} g_1 H &= \{(1\,3), (1\,3)(1\,2)\} = \{(1\,3), (1\,2\,3)\} \\ g_2 H &= \{(1\,2\,3), (1\,2\,3)(1\,2)\} = \{(1\,2\,3), (1\,3)\} \\ g_3 H &= \{(1\,3\,2), (1\,3\,2)(1\,2)\} = \{(1\,3\,2), (2\,3)\} \end{aligned}$$

We observe that $g_1 H = g_2 H$. This is not surprising. Afterall, $g_2^{-1} g_1 = (1\,2\,3)^{-1}(1\,3) = (1\,3\,2)(1\,3) = (1\,2) \in H$. Further, we note that $S_3 = H \sqcup g_1 H \sqcup g_3 H$. Thus, $A = \{1, g_1, g_3\} = \{1, (1\,3), (1\,3\,2)\}$ is a subset of S_3 as asserted in Theorem 12.2. This subset is not unique to the needs of that theorem, as $B = \{(1\,2), (1\,2\,3), (1\,3\,2)\}$ would have served the same purpose.

Let us take one more example: $\mathrm{rot}(P_n) \leq \mathrm{sym}(P_n)$. Revisit our earlier discussion on rotation and reflection symmetries of regular polygons, and conclude that $\mathrm{sym}(P_n) = \mathrm{rot}(P_n) \sqcup f\mathrm{rot}(P_n)$, where f is a reflection symmetry of P_n.

Thus, the notion of cosets allows us to equipartition a group G into subsets, each one of which is of the same size as a given subgroup H. This will helps us establish Lagrange's Theorem in the next lecture.

Lecture 13

Cosets : Left and Right

February 20, 2025

We continue the setup of the last lecture: we are given a group G and a subgroup H of G. We understand that we can translate H by element $g \in G$ to obtain non-overlapping sets $gH := \{gh : h \in H\}$, called *cosets* of H in G. These sets are in bijection with each other. We also determined these cosets for a few examples.

A group is called *finite* if it contains only finitely many elements. The number of elements in a group G is denoted by $|G|$. Thus, for a finite group G we have $|G| < \infty$.

13.1. Lagrange's Theorem

Theorem 13.1. *Let G be a finite group, and H be a subgroup of G. Then $|H|$ divides $|G|$.*

Proof. In the last lecture we partitioned G into cosets of equal parts. That is precisely the idea that goes into this proof. Recall Theorem 12.2. Let $A := \{g_1, g_2, \ldots, g_r\} \subseteq G$ be such that $G = g_1H \sqcup g_2H \cdots \sqcup g_rH$ is a disjoint union of cosets that covers G. Then,

$$|G| = |g_1H| + |g_2H| + \cdots + |g_rH|$$

Since each g_iH is in bijection with H, we have

$$|G| = \underbrace{|H| + |H| + \cdots + |H|}_{r \text{ times}} = r|H|$$

Thus, $|H|$ divides $|G|$. $\square$

We see from the proof that the quotient $|G|/|H|$ is equal r, which is the number of distinct cosets of H in G. The number of distinct cosets of H is G is called the *index* of H in G. It is denoted by $[G : H]$.

13.2. Right cosets

The cosets that we have considered so far are *left* cosets, as these are obtained by translating elements of H on left by an element $g \in G$. What difference would it have made, had we done this translation on right? To proceed, let us define

$$Hg := \{hg : h \in H\}$$

Let us call it a *right coset* of H in G, represented by g. What is the first natural question? Is $gH = Hg$? It is clear that if $g = 1 \in G$, then $gH = Hg = H$. What about other $g \in G$? Let us revisit the same example.

Example 13.2 We take $G = S_3$ and $H = \{1, (1\ 2)\}$. Then, for $g = (2\ 3)$, we calculate

$$gH = \{(2\ 3), (2\ 3)(1\ 2)\} = \{(2\ 3), (1\ 3\ 2)\}$$
$$Hg = \{(2\ 3), (1\ 2)(2\ 3)\} = \{(2\ 3), (1\ 2\ 3)\}$$

So, we have our answer. In general, $gH \neq Hg$. However, convince yourself that we would have done everything that we did in the last lecture, using right cosets. Proposition 12.1 and Theorem 12.2 would stand equally firm, and the proof of Lagrange's Theorem would have gone through essentially verbatim. Further, the index of H in G does not change when we replace, in our calculation, left cosets by right cosets.

Thus, conceptually, it makes no difference if we prefer left cosets over right cosets. Having said that, let us make an interesting observation.

Exercise 13.3 Let G be a finite group and H be a subgroup of G such that $|H| = \frac{1}{2}|G|$. Then $gH = Hg$ for every $g \in G$.

How to go about it? Since $|G|/|H| = 2$, the proof of Lagrange's Theorem suggests that there are only two left cosets: H and its complement $G \backslash H$. Rewriting the proof of Lagrange's Theorem using right cosets would also suggest that there are only two right cosets: H and its complement $G \backslash H$. Thus, for $g \in H$ we have $gH = H = hg$, and for $g \in G \backslash H$ we have $gH = G \backslash H = Hg$.

A subgroup H of G is called a *normal subgroup* if for every $g \in G$, we have $gH = Hg$. The subgroup $H = \{1, (1\ 2)\}$ of S_3 is not a normal subgroup. If $H \leq G$ is such that $[G : H] = 2$, then H is a normal subgroup of G. A group may contain normal subgroups in addition to the subgroups of index 2. For example, if G is abelian then all subgroups of G are normal. We write $H \trianglelefteq G$ to express that H is a normal subgroup of G.

13.3. The quotient G/H

Let us denote the set of all left cosets of H in G by G/H. Thus,

$$G/H := \{gH : g \in G\}$$

If G is finite then how many elements does G/H have? It is exactly $[G : H]$. If you revisit Theorem 12.2, then it is same as the size of the set A.

Let us aspire to infuse the group structure of G on the set G/H. What is the natural thing that comes to your mind? For $g_1H, g_2H \in G/H$, let us define

$$g_1H \star g_2H = g_1g_2H \in G/H$$

Unfortunately, this is not a good definition. Let alone good, it is not even a definition! This is because the dependence of the right hand side on the choice of g_1 and g_2 is not coherent with the dependence of the choice of g_1 and g_2 on the left hand side. Let us exhibit this point with the help of an example. For $G = S_3$ and $H = \{1, (1\ 2)\}$, observe that

$$(1\ 3)H = (1\ 2\ 3)H$$

but

$$(2\ 3)H = (1\ 3)(1\ 3\ 2)H = (1\ 3)H \star (1\ 3\ 2)H = (1\ 2\ 3)H \star (1\ 3\ 2)H = (1\ 2\ 3)(1\ 3\ 2)H = H$$

dumps us into a contradiction $(2\ 3)H = H$. This happened because we trusted the $\star$ for a valid definition. The following Theorem compensates us for this disappointment.

Theorem 13.4. *Let $H \trianglelefteq G$ (that is H is a normal subgroup of G). Let G/H be the set of left cosets of H in G. Then $\star$ defined by $g_1H \star g_2H = g_1g_2H$ is a composition law that make $(G/H, \star)$ into a group.*

The group $(G/H, \star)$, when $H \trianglelefteq G$, is called the *quotient of G by H*.

Let us exhibit examples. Recall that A_n, the group of even permutations, is a subgroup of index 2 in S_n. Hence, $A_n \trianglelefteq S_n$. The quotient group is:

$$S_n/A_n = \{A_n, (1\ 2)A_n\}$$

where $\star$ as in Theorem 13.4 can be written in the tabular form as follows.

$\star$	A_n	$(1\ 2)A_n$
A_n	A_n	$(1\ 2)A_n$
$(1\ 2)A_n$	$(1\ 2)A_n$	A_n

Let us write down examples of groups of size 2.

Example 13.5

1. S_n/A_n.
2. S_2, the group of permutation of two symbols.
3. $\{1, -1\}$ under ordinary multiplication of numbers.
4. $\{0, 1\}$, under addition in Boolean arithmetic: $0+0 = 0, 0+1 = 1, 1+0 = 1, 1+1 = 0$. In logic, this corresponds to the operation XOR (exclusive or), whose truth table is as follows.

XOR	F	T
F	F	T
T	T	F

5. $\{R_0, R_\pi\}$, where R_θ denotes anti-clockwise rotation in $\mathbb{R}^2$ by angle θ.

6. $\{R_0, F\}$, where F is a reflection in $\mathbb{R}^2$ about a line passing through origin.

Is there a *fundamental similarity* in these examples? Yes! While they are born out of different contexts, at core, all of them are manifestations of the group expressed through the following composition table.

$$\begin{array}{c|cc} & \circ & \bullet \\ \hline \circ & \circ & \bullet \\ \bullet & \bullet & \circ \end{array}$$

In that case, all these groups are *isomorphic*–of the same form.

In the next section, we discuss the concept of homomorphism. The concept of isomorphism will be discussed in the next lecture.

13.4. Homomorphisms: *of the similar form*

Let G_1 and G_2 be two groups. A function $f : G_1 \to G_2$ is called a *homomorphism* if the following holds for every $x, y \in G_1$: $f(xy) = f(x)f(y)$. Here, the composition on the left hand side is that of G_1, and the one on the right hand side is that of G_2.

In highschool, we saw examples of homomorphism while learning exponentiation and logarithm. We revisit it now in the language of groups.

The groups are $(\mathbb{R}, +)$, the group of real numbers under addition; and $(\mathbb{R}^{>0}, \times)$, the group of positive reals under multiplication. Define the map

$$\exp_2 : (\mathbb{R}, +) \to (\mathbb{R}^{>0}, \times)$$

by $\exp_2(x) := 2^x$. Then $\exp_2(x + y) = 2^{x+y} = 2^x 2^y = \exp_2(x)\exp_2(y)$. Hence $x \overset{\exp_2}{\longmapsto} 2^x$ is a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}^{>0}, \times)$.

There is a homomorphism in the reverse direction too:

$$\log_2 : (\mathbb{R}^{>0}, \times) \to (\mathbb{R}, +)$$

which is given by $x \longmapsto \log_2(x)$. Here, $\log_2(x)$ is the logarithm of x at the base 2. The fact that $\log_2(xy) = \log_2(x) + \log_2(y)$ confirms that $\log_2$ is a homomorphism. We will discuss more on these ideas in the next lecture.

Lecture 14

Group Actions

February 21, 2025

14.1. Homomorphisms continued...

Recall, from the last lecture, that a homomorphism is a function $f : G_1 \to G_2$ that satisfies $f(xy) = f(x)f(y)$ for every $x, y \in G_1$. Substituting $y = 1_{G_1}$ we obtain

$$f(y) = f(1_{G_1}y) = f(1_{G_1})f(y),$$

for every $y \in G_1$. Therefore, $f(1_{G_1}) = f(y)f(y)^{-1} = 1_{G_2}$, that is, a homomorphism maps the identity element of G_1 to the identity element of G_2.

A trivial example of homomorphism is the function $f : G_1 \to G_2$ that maps every element of G_1 to the identity of G_2. Let us see more interesting examples.

1. Let G denote the subgroup of $\mathrm{sym}(\mathbb{R}^2)$ generated by reflections and rotations. Then the map $f : (\mathbb{Z}, +) \to G$ defined by $f(r) = R_{2\pi r/\sqrt{2}}$ is a homomorphism.
2. The map $f : (\mathbb{Z}, +) \to \mathrm{rot}(P_n)$ defined by $f(r) = R_{2\pi r/n}$ is a homomorphism.
3. The signature map $\mathrm{sgn} : S_n \to \{1, -1\}$ that map a permutation σ to its signature, is a homomorphism.
4. Let $\mathrm{GL}_n(\mathbb{R})$ denote the group of $n \times n$ invertible matrices under matrix multiplication, and $R^\times$ denote the multiplicative group of nonzero real numbers. Then the determinant map $\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times$ is a homomorphism.
5. Let G be a group and H be a normal subgroup of G. Let $(G/H, \star)$ denote the quotient of G by H, as in Theorem 3. Then the map $f : G \to (G/H, \star)$ defined by $f(g) = gH$ is a homomorphism.

A homomorphism $f : G_1 \to G_2$ is called an *isomorphism*, if it is bijective. The function $f : S_3 \to \mathrm{sym}(P_3)$ defined below is an example of isomorphism.

$$f(1) = 1, \quad f((1\ 2)) = f_3, \quad f((1\ 3)) = f_2, \quad f((2\ 3)) = f_1$$
$$f((1\ 2\ 3)) = r_{2\pi/3}, \quad f((1\ 3\ 2)) = r_{4\pi/3}$$

Two groups G_1 and G_2 are called *isomorphic*, if there exists an isomorphism from G_1 to G_2. Any two group of size 2 are isomorphic, though the same does not hold for any two groups of size 4.

14.2. Group Actions

The reason groups are interesting and useful is because they act! Let G be a group and X be a set. An *action of G on X* is a map

$$\diamond : G \times X \to X$$

that maps (g, x) to $g \diamond x$, such that

(i). $1 \diamond x = x$ for all $x \in X$.

(ii). $g \diamond (h \diamond x) = gh \diamond x$ for all $x \in X$ and all $g, h \in G$. Here gh is the composition of g and h in the group G.

Example 14.1 Here are some examples of group actions.

(i). Let S be a shape in $\mathbb{R}^2$ or $\mathbb{R}^3$, and $\operatorname{sym}(S)$ be the group of symmetries of S. Then $\operatorname{sym}(S)$ acts naturally on S as follows:

$$\sigma \diamond x = \sigma(x)$$

for $\sigma \in \operatorname{sym}(S)$ and $x \in X$.

(ii). The group S_n of permutations on n symbols acts on the set $\{1, 2, \ldots, n\}$ as follows:

$$\sigma \diamond i = \sigma(i)$$

for $\sigma \in S_n$ and $i \in \{1, 2, \ldots, n\}$.

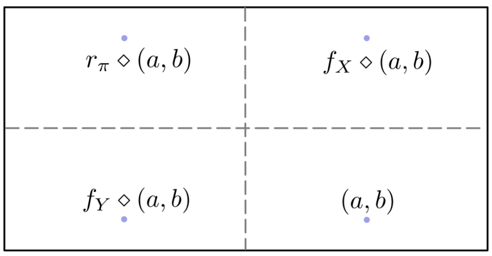
(iii). Consider the group $\operatorname{sym}(R) := \{1, f_X, f_Y, r_\pi\}$ consisting of symmetries of a rectangle. Let

$$X := \{(a, b) : |a| \leq 2, |b| \leq 1\}$$

Then the map $\diamond : \operatorname{sym}(R) \times X \to X$ defined by

$$\begin{aligned} 1 \diamond (a, b) &= (a, b) \\ f_X \diamond (a, b) &= (a, -b) \\ f_Y \diamond (a, b) &= (-a, b) \\ r_\pi \diamond (a, b) &= (-a, -b) \end{aligned}$$

is a group action. The following depicts this action pictorially.



Lecture 15

Orbits

February 24, 2025

Imagine tetrahedron as a toy. Lets make it colourful! You have two colours: red and blue. You are allowed to paint each face of the tetrahedron in either of these colours. How many such colourful tetrahedral toys are possible? Since for each face there are two choices, the total number of possibilities is $2^4 = 16$. Let us name these colourful tetrahedrons. The one for which all faces are coloured blue will be named (B, B, B, B), and the one for which second and fourth faces are colored blue and other faces are coloured red will be named (R, B, R, B).

Let us place these 16 coloured toys on a table. Perform a nontrivial rotation on one of these. Do you see this rotated toy now appears same as one of the rest 15? Hence, if the rotational symmetry is to be accounted, the number of colored toys is actually much less than 16. This is because any two toys that can be converted from one to the other by means of a rotational symmetry of tetrahedron, are to be considered the same. In other words, they are in the same *rotational orbit*. This combinatorics using symmetry considerations can be worked out using group action. But, where is the group action here? Let us rewrite this story through the following example of group action.

Example 15.1 Take a tetrahedron with its faces labelled $1, 2, 3, 4$. Then the group of rotational symmetries of this tetrahedron is A_4 – the group of permutations of 4 faces by means of an even permutation. Let $X = \{(c_1, c_2, c_3, c_4) : c_i = R \text{ or } B\}$. Do you see X as the set of all 16 coloured tetrahedral toys, without symmetry consideration? For $\sigma \in A_4$ and (c_1, c_2, c_3, c_4), define

$$\sigma \diamond (c_1, c_2, c_3, c_4) = (c_{\sigma(1)}, c_{\sigma(2)}, c_{\sigma(3)}, c_{\sigma(4)})$$

This is how coloured toy (c_1, c_2, c_3, c_4) converts to another one $(c_{\sigma(1)}, c_{\sigma(2)}, c_{\sigma(3)}, c_{\sigma(4)})$. via rotation defined by σ. Thus, $(1\ 2)(3\ 4) \diamond (R, B, R, B) = (B, R, B, R)$. The map

$$A_4 \times X \to X$$

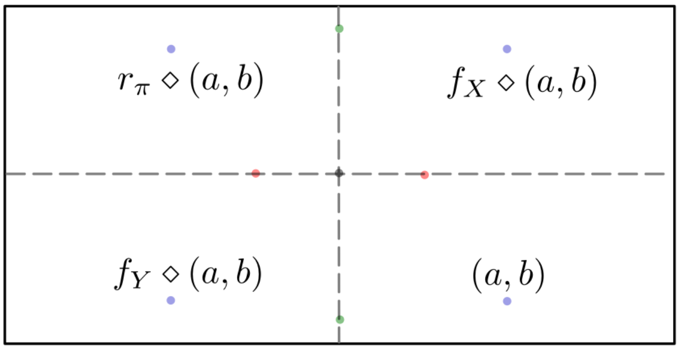
defined by $(\sigma, x) \mapsto \sigma \diamond x$ is an action of A_4 on X.

A given toy $x \in X$ can be converted, by means of a rotation, to the toys occurring in the set $\{\sigma \diamond x : \sigma \in A_4\}$. This set is called the *orbit* of x under the action $\diamond$. The question of counting distinct coloured toys is equivalent to the question of counting number of distinct orbits for this action.

Definition 15.2. *Let $\diamond : G \times X \to X$ be a group action. Let $x \in X$. Then the* orbit *of x is the following subset of X:*

$$\text{orbit}(x) := \{g \diamond x : g \in G\}$$

In Example 14.1(iii), points with the same colour are in the same orbit.



In Example 14.1(ii), where the group S_n of permutations acts on n symbols $\{1, 2, \ldots, n\}$, the $\text{orbit}(1) = \{1, 2, \ldots, n\}$. This is because 1 can be moved to any symbol by means of a suitable permutation. In fact, $\text{orbit}(i) = \{1, 2, \ldots, n\}$ for any $i \in \{1, 2, \ldots, n\}$. Thus, this action has only one orbit.

Let $\diamond : G \times X \to X$ be an action. Orbits partition X into disjoint subsets, as we see from the following proposition.

Proposition 15.3. *Let $\diamond : G \times X \to X$ be a group action. Let $x, y \in X$, and $\text{orbit}(x)$ and $\text{orbit}(y)$ be their respective orbits. Then, either $\text{orbit}(x) \cap \text{orbit}(y) = \emptyset$, or $\text{orbit}(x) = \text{orbit}(y)$.*

Proof. This proposition is strikingly similar to the partition of a group into cosets. Suppose, $\text{orbit}(x) \cap \text{orbit}(y) = \emptyset$. Then there is nothing to show. Hence, we assume that the intersection of two orbits is nonempty. $z \in \text{orbit}(x) \cap \text{orbit}(y)$. Then

$$z = g_1 \diamond x = g_2 \diamond y$$

for suitable $g_1, g_2 \in G$. Therefore,

$$g_2^{-1} \diamond (g_1 \diamond x) = g_2^{-1} \diamond (g_2 \diamond y) = g_2^{-1} g_2 \diamond y = 1 \diamond y = y$$

Hence $y = g_2^{-1} g_1 \diamond x \in \text{orbit}(x)$. Further, for an arbitrary $g \in G$, we have

$$g \diamond y = g g_2^{-1} g_1 \diamond x \in \text{orbit}(x)$$

Thus, $\text{orbit}(y) \subseteq \text{orbit}(x)$. A similar argument shows that $\text{orbit}(x) \subseteq \text{orbit}(y)$. Hence, $\text{orbit}(x) = \text{orbit}(y)$. □

Lecture 16

Stabilizer and fixed points

March 03, 2025

In this lecture, we introduce the notion of stabilizers and fixed point for a group action $\diamond : G \times X \to X$.

16.1. Stabilizer

Let $x \in X$, the *stabilizer* of x is defined by

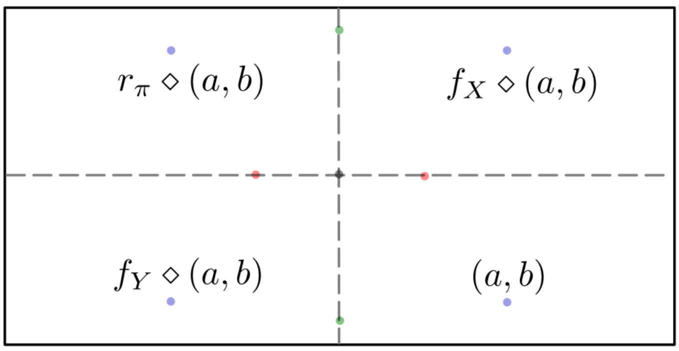
$$\mathrm{stab}(x) := \{g \in G : g \diamond x = x\}$$

Thus, the set of elements of G that do not actively participate in moving a point $x \in X$, is called stabilizer of x. Stabilizers are subgroups of G, for the following reasons.

1. If $g, h \in G$ are such that $h \diamond x = x$ and $g \diamond x = x$, then
$$gh \diamond x = g \diamond (h \diamond x) = g \diamond x = x$$

2. If $g \in G$ is such that $g \diamond x = x$, then
$$g^{-1} \diamond x = g^{-1} \diamond (g \diamond x) = g^{-1} g \diamond x = x$$

Let us revisit Example 14.1 (iii).



In this example, the stabilizer of points coloured blue is $\{1\}$, the stabilizer of points coloured red is $\{1, f_X\}$, and the points coloured green is $\{1, f_Y\}$. The stabilizer of the origin, which is coloured black, is the entire group $\{1, f_X, f_Y, r_\pi\}$. We note that the subgroup $\{1, r_\pi\}$ of the group $\mathrm{sym}(R)$ does not occur as stabilizer in this group action. Thus, stabilizers are subgroups of the group that participates in an action, but not all subgroups occur as an stabilizer.

From this example, we observe that an element with a larger orbit has a smaller stabilizer. The following theorem beautifully expresses this point.

Theorem 16.1 (Orbit-stabilizer Theorem)**.** *Let G be a finite group and $\diamond : G \times X \to X$ be an action. Let $x \in X$. Then $|\mathrm{orbit}(x)| \times |\mathrm{stab}(x)| = |G|$.*

The proof is based on the observation that the set of cosets of $\mathrm{stab}(x)$ in G is in bijection with $\mathrm{orbit}(x)$. We postpone details for a later lecture.

16.2. Fixed Points

In the definition of stabilizer, we focused on an element $x \in X$ and ask which elements of the group G do not move x. We turn the table, and focus on an element $g \in G$. We then ask, which elements of X are not moved by this g. These elements are called *fixed points* of g.

More formally, let $\diamond : G \times X \to X$ be a group action. Let $g \in G$. We define the set of *fixed points* of g as follows:

$$\mathrm{fixed}(g) := \{x \in X : g \diamond x = x\}$$

In Example 14.1 (ii), with $n = 4$, take $\sigma = (1\ 2) \in S_4$ and $i = 1$. Then we have

(i). $\mathrm{fixed}(\sigma) = \{3, 4\}$.

(ii). $\mathrm{orbit}(1) = \{1, 2, 3, 4\}$.

(iii). $\mathrm{stab}(1) = \{1, (2\ 3), (2\ 4), (3\ 4), (2\ 3\ 4), (2\ 4\ 3)\}$.

In this example, we can verify orbit-stabilizer theorem:

$$4 \times 6 = |\mathrm{orbit}(1)| \times |\mathrm{stab}(1)| = |S_4| = 24$$

Lecture 17

Counting orbits

March 17, 2025

In this lecture, we will derive a powerful result, called *Burnside's Lemma* to count the number of orbits for an action of a finite group on finite sets. Setting aside the fact that the lemma is not originally due to Burnside, the result expresses the number of orbits in terms of fixed points of the group action.

Lemma 17.1. *Let G be a finite group and $\diamond : G \times X \to X$ be an action, with X finite. Let n be the number of orbits for this action. Then,*

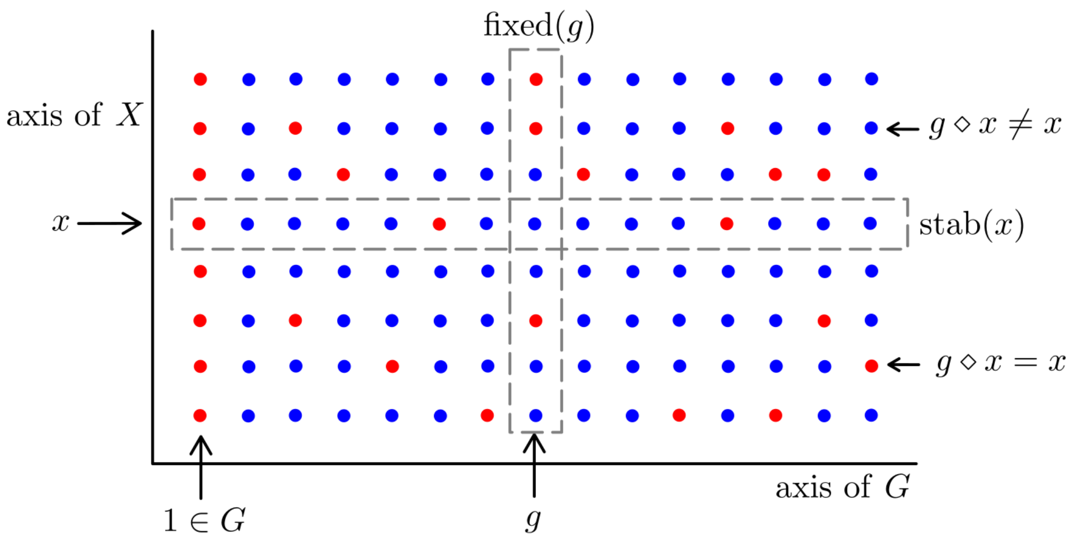
$$n = \frac{1}{|G|} \sum_{g \in G} |\text{fixed}(g)|$$

Thus, to restate it, average number of fixed points of an action is equal to the number of its orbits.

Proof. We arrange the elements of $G \times X$ as a grid and consider the subset

$$\mathcal{F} := \{(g, x) : g \diamond x = x\}$$

In the picture, the elements of $\mathcal{F}$ are depicted in red colour. Since $1 \in G$ fixes all points



of X, it is clear that at least one column of points will be entirely red. We count red points of $G \times X$ in two different ways.

Counting columnwise. For each $g \in G$, the number of red points in the corresponding column is equal to $|\text{fixed}(g)|$. Thus,

$$|\mathcal{F}| = \sum_{g \in G} |\text{fixed}(g)|$$

Counting rowwise. For $x \in X$, a column corresponding to g contributes to a red point precisely when $\text{stab}(x)$. Thus, the number of red points in the row corresponding to x is equal to $|\text{stab}(x)|$, and

$$|\mathcal{F}| = \sum_{x \in X} |\text{stab}(x)|$$

From the orbit-stabilizer theorem, for each $x \in X$ we have $|\text{orbit}(x)| \times |\text{stab}(x)| = |G|$. Substituting this, we obtain

$$|\mathcal{F}| = \sum_{x \in X} \frac{|G|}{|\text{orbit}(x)|} = |G| \sum_{x \in X} \frac{1}{|\text{orbit}(x)|}$$

In the expression $\sum_{x \in X} \frac{1}{|\text{orbit}(x)|}$, the contribution from each $x \in X$ is $\frac{1}{|\text{orbit}(x)|}$. Thus, the collective contribution of the entire orbit of x in this expression is

$$|\text{orbit}(x)| \times \frac{1}{|\text{orbit}(x)|} = 1.$$

In other words, each orbit adds 1 to this expression. That is, $\sum_{x \in X} \frac{1}{|\text{orbit}(x)|} = n$, the number of orbits for the action. Now comparing the number of red points through these two count we obtain

$$|\mathcal{F}| = \sum_{g \in G} |\text{fixed}(g)| = |G| \sum_{x \in X} \frac{1}{|\text{orbit}(x)|} = |G| \times n$$

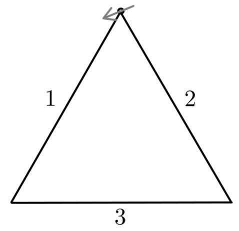
That is,

$$n = \frac{1}{|G|} \sum_{g \in G} |\text{fixed}(g)|.$$

□

Let us use Burnside's lemma for a counting problem.

Example 17.2 Using a pallette of s distinct colours, three sides of a frame in the shape of an equilateral triangle are to be painted. How many distinct coloured frames can be formed?



To start with, let us fix such a frame on a wall.

For each of the three sides, we have s choices of colors. Thus, we can form s^3 wall mounted coloured frames. Let us denote the set of wall mounted coloured frames by X. A typical element of X is denoted by (c_1, c_2, c_3), where c_i denotes a colour available in the given pallette. We realize there is redundancy in X if we take these frames off the wall and rotate or flip them. Many frames that looked different on the wall appear the same once they are off the wall. That's where a group action comes into picture. What is the relevant group here, and what is the action?

The group that acts is the group of symmetries of an equilateral triangle,

$$\mathrm{sym}(\Delta) = \{1, r, r^2, f_1, f_2, f_3\}$$

The group action $\diamond : \mathrm{sym}(\Delta) \times X \to X$ is given by

$$\sigma \diamond (c_1, c_2, c_3) = (c_{\sigma(1)}, c_{\sigma(2)}, c_{\sigma(3)})$$

Two wall mounted frame (c_1, c_2, c_3) and (d_1, d_2, d_3) are the same if and only if for some $\sigma \in \mathrm{sym}(\Delta)$, we have

$$(d_1, d_2, d_3) = (c_{\sigma(1)}, c_{\sigma(2)}, c_{\sigma(3)})$$

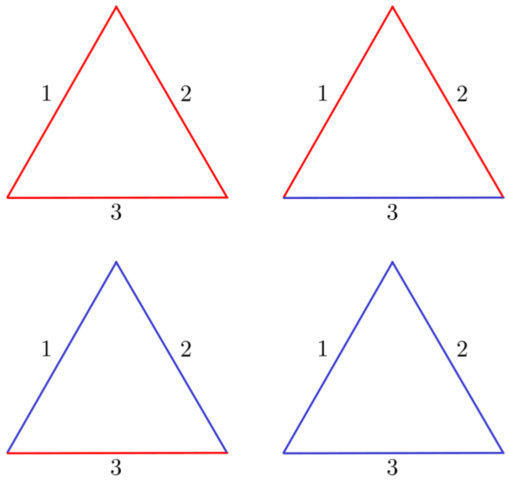
That is, (c_1, c_2, c_3) and (d_1, d_2, d_3) are in the same orbit. Thus, the count of distinct coloured frames is equal to the number of orbits for this action. By Burnside's formula, this is equal to

$$n = \frac{1}{|\mathrm{sym}(\Delta)|} \sum_{\sigma \in \mathrm{sym}(\Delta)} |\mathrm{fixed}(\sigma)|$$

Note that if σ is a nontrivial rotation then $(c_1, c_2, c_3) \in \mathrm{fixed}(\sigma)$ if and only if $c_1 = c_2 = c_3$, and if σ is a reflection then $(c_1, c_2, c_3) \in \mathrm{fixed}(\sigma)$ if and only if exactly two of c_1, c_2, c_3 are equal and the other color is different from these two. With this observation, we calculate the number of fixed points in the following table.

σ	$\lvert\mathrm{fixed}(\sigma)\rvert$
1	s^3
r	s
r^2	s
f_1	s^2
f_2	s^2
f_3	s^2
n	$\frac{1}{6}s(s+1)(s+2)$

Thus, the total number of distinct coloured frames is $\frac{1}{6}s(s+1)(s+2)$. When there are two colours in our pallette, the number of such frames is 4.



Lecture 18

Conjugation action

March 19, 2025

One of the most important actions in the theory of groups is conjugation action. Let G be a group. For the conjugation action, the set X is G itself, treated merely as a set. The action $\diamond : G \times G \to G$ is given by $g \diamond x = gxg^{-1}$. Since $1 \diamond x = 1x1^{-1} = x$, and

$$\begin{aligned} g \diamond h \diamond x &= g \diamond hxh^{-1} \\ &= ghxh^{-1}g^{-1} \\ &= ghx(gh)^{-1} \\ &= gh \diamond x \end{aligned}$$

$\diamond$ is indeed a group action. Orbits under this action are called *conjugacy classes*. Elements in the same conjugacy class are called *conjugates* of each other. Thus, two elements $x, y \in G$ *conjugates* if there exists $g \in G$ such that $y = gxg^{-1}$. Let G be a finite group and $\mathcal{C}_1, \mathcal{C}_2, \cdots, \mathcal{C}_s$ be all its conjugacy classes. Then,

$$G = \mathcal{C}_1 \sqcup \mathcal{C}_2 \sqcup \cdots \sqcup \mathcal{C}_s$$

is a disjoint union. The equation

$$|G| = |\mathcal{C}_1| + |\mathcal{C}_2| + \cdots + |\mathcal{C}_s|$$

is called the *class equation* of G. For the conjugation action, $\operatorname{orbit}(1) = \{1\}$, where $1 \in G$. Thus, the integer 1 necessarily appears on the right hand side of the class equation. We observe that for an abelian group G, every conjugacy class is singleton. Thus, for abelian groups the class equation is

$$|G| = \underbrace{1 + 1 + \cdots + 1}_{|G| \text{ times}}$$

In fact, if the class equation of a group G has only $1s$, then the group is necessarily abelian. Thus, from the class equation one may hope to extract some information about the group.

18.1. Conjugacy classes in S_3

We start with some conjugacy calculations on $S_3 = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$.

$$(2\ 3)(1\ 2)(2\ 3)^{-1} = (2\ 3)(1\ 2)(2\ 3) = (1\ 3)$$
$$(1\ 3)(1\ 2)(1\ 3)^{-1} = (1\ 3)(1\ 2)(1\ 3) = (2\ 3)$$
$$(1\ 2)(1\ 2\ 3)(1\ 2)^{-1} = (1\ 2)(1\ 2\ 3)(1\ 2) = (1\ 3\ 2)$$
$$(1\ 3)(1\ 2\ 3)(1\ 3)^{-1} = (1\ 3)(1\ 2\ 3)(1\ 3) = (1\ 3\ 2)$$
$$(2\ 3)(1\ 2\ 3)(2\ 3)^{-1} = (2\ 3)(1\ 2\ 3)(2\ 3) = (1\ 3\ 2)$$

These calculation suggest that there are three conjugacy classes in S_3.

$$\mathcal{C}_1 = \{1\}, \quad \mathcal{C}_2 = \{(1\ 2), (1\ 3), (2\ 3)\}, \quad \mathcal{C}_3 = \{(1\ 2\ 3), (1\ 3\ 2)\}$$

and the class equation of S_3 is

$$|S_3| = 6 = 1 + 2 + 3$$

Since the number of conjugacy classes is precisely the number orbits under a specific action, namely the conjugation action, we can use Burnside's formula as well, to count the number of conjugacy classes. The following table will help with this calculation.

σ	fixed(σ)	$\lvert$fixed$(\sigma)\rvert$
1	S_3	6
$(1\ 2)$	$\{1, (1\ 2)\}$	2
$(1\ 3)$	$\{1, (1\ 3)\}$	2
$(2\ 3)$	$\{1, (2\ 3)\}$	2
$(1\ 2\ 3)$	$\{1, (1\ 2\ 3), (1\ 3\ 2)\}$	3
$(1\ 3\ 2)$	$\{1, (1\ 2\ 3), (1\ 3\ 2)\}$	3
		$\Sigma = 18$

Therefore, the number of conjugacy classes in S_3 is $\frac{18}{6} = 3$.

What about S_n, for arbitrary n?

18.2. Conjugacy classes in S_n

For $n \in \mathbb{N}$, a tuple $\overline{r} = (r_1, r_2, \dots r_s)$, where each $r_i \in \mathbb{N}$, is called a *partition* of n if

(i). $r_1 \geq r_2 \geq \cdots \geq r_s$, and

(ii). $r_1 + r_2 + \cdots + r_s = n$.

We write $\overline{r} \vdash n$ to express that $\overline{r} = (r_1, r_2, \dots r_s)$ is a partition of n. The following is the set of all partitions of 3.

$$\mathcal{P}(3) := \{(3), (2, 1), (1, 1, 1)\}$$

Similarly, for $n = 4$ and 5, following are the sets of partitions.

$$\mathcal{P}(4) := \{(4), (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1)\}$$

$$\mathcal{P}(5) := \{(5), (4, 1), (3, 2), (3, 1, 1), (2, 2, 1), (2, 1, 1, 1), (1, 1, 1, 1, 1)\}$$

The function $n \mapsto p(n) := |\mathcal{P}(n)|$ is call the *partition function*. Clearly, $p(2) = 2, p(3) = 3, p(4) = 5, p(5) = 7$.

Here, our aim is to establish the that the number of conjugacy classes in S_n is equal to $p(n)$. In fact, there is a bijective correspondence between conjugacy classes of S_n and $\mathcal{P}(n)$.

Lemma 18.1. *Let $[1..n] := \{1, 2, \ldots, n\}$ and $r \leq n$. Let $a_1, a_2, \ldots, a_r$ be distinct elements in $[1..n]$, and $\tau = (a_1\ a_2\ \ldots\ a_r) \in S_n$. Then, $\sigma\tau\sigma^{-1} = (\sigma(a_1)\ \sigma(a_2)\ \ldots\ \sigma(a_r))$ for every $\sigma \in S_n$.*

Proof. Let us see what $\sigma\tau\sigma^{-1}$ does to $\sigma(a_1)$.

$$\sigma\tau\sigma^{-1}(\sigma(a_1)) = \sigma\tau(a_1) = \sigma(a_2)$$

In fact, for each $i < r$,

$$\sigma\tau\sigma^{-1}(\sigma(a_i)) = \sigma\tau(a_i) = \sigma(a_{i+1})$$

and

$$\sigma\tau\sigma^{-1}(\sigma(a_r)) = \sigma\tau(a_r) = \sigma(a_1)$$

Further, for $k \notin \{\sigma(a_1), \ldots, \sigma(a_r)\}$, let $\ell = \sigma^{-1}(k)$. Then $\ell \notin \{a_1, \ldots, a_r\}$, so $\tau(\ell) = \ell$. Thus,

$$\sigma\tau\sigma^{-1}(k) = \sigma\tau(\ell) = \sigma(\ell) = k,$$

Clearly, the cycle $(\sigma(a_1)\ \sigma(a_2)\ \ldots\ \sigma(a_r))$ also fixes k. Hence, $\sigma\tau\sigma^{-1} = (\sigma(a_1)\ \ldots\ \sigma(a_r))$, as they agree on all elements of $[1..n]$. □

The Lemma 18.1 states that a conjugate of an r-cycle is again an r-cycle. Conversely, any two r-cycles are conjugate to each other.

Lemma 18.2. *Let $\tau = (a_1\ a_2\ \ldots\ a_r)$ and $\tau' = (b_1\ b_2\ \ldots\ b_r)$ be two r-cycles in S_n. Then there exists $\sigma \in S_n$ such that $\sigma\tau\sigma^{-1} = \tau'$.*

Proof. We draw inspiration from the proof of Lemma 18.1. Let σ be such that $\sigma(a_i) = b_i$, and it fixes symbols not appearing in τ. Then

$$\sigma\tau\sigma^{-1} = (\sigma(a_1)\ \sigma(a_2)\ \ldots\ \sigma(a_r)) = (b_1\ b_2\ \ldots\ b_r) = \tau'.$$

□

Any permutation τ can be written as

$$\tau = (a_{11}\ a_{12}\ \ldots\ a_{1r_1})(a_{21}\ \ldots\ a_{2r_2})(a_{31}\ \ldots\ a_{3r_3})\ldots(a_{s1}\ \ldots\ a_{sr_s}) \in S_n$$

where $\{a_{11}, \ldots, a_{1r_1}, a_{21}, \ldots, a_{2r_2}, \ldots, a_{s1}, \ldots, a_{sr_s}\} = [1..n]$; $r_1 \geq r_2 \geq \cdots \geq r_s$; and $r_1 + r_2 + \cdots + r_s = n$. The tuple $\overline{r} := (r_1, r_2, \ldots, r_s)$ is a partition. It is called the *partition type* of τ.

For example, the 3-cycles in S_5 can be written as $(a_{11}\ a_{12}\ a_{13})(a_{21})(a_{31})$. Thus, the partition type of 3-cycles in S_5 is $(3, 1, 1)$. The partition type of 3-cycles in S_7 will be $(3, 1, 1, 1, 1)$.

Theorem 18.3. *Two permutation τ and τ' of S_n are in the same orbit with respect to conjugation action of S_n if and only if they have the same partition type.*

Proof. Let $\overline{r} := (r_1, r_2, \ldots, r_s)$ be the partition type of τ in S_n. We write

$$\tau = (a_{11}\ a_{12}\ \ldots\ a_{1r_1})(a_{21}\ \ldots\ a_{2r_2})(a_{31}\ \ldots\ a_{3r_3})\ldots(a_{s1}\ \ldots\ a_{sr_s})$$

The proof follows from Lemma 18.1 and Lemma 18.2, and the fact that for any $\sigma \in S_n$,

$$\sigma\tau\sigma^{-1} = \sigma(a_{11}\ a_{12}\ \ldots\ a_{1r_1})\sigma^{-1}\sigma(a_{21}\ \ldots\ a_{2r_2})\sigma^{-1}\sigma(a_{31}\ \ldots\ a_{3r_3})\sigma^{-1}\ldots\sigma(a_{s1}\ \ldots\ a_{sr_s})\sigma^{-1}$$

This theorem establishes one to one correspondence between conjugacy classes in S_n and partitions of n. The following table does the detailed analysis of conjugacy classes for S_5, for each partition of 5.

$$\mathcal{P}(5) := \{(5), (4, 1), (3, 2), (3, 1, 1), (2, 2, 1), (2, 1, 1, 1), (1, 1, 1, 1, 1)\}$$

$\overline{r} \in \mathcal{P}(5)$	Conjugacy class $\mathcal{C}$	$\lvert\mathcal{C}\rvert$
(5)	orbit$((1\ 2\ 3\ 4\ 5))$	$4! = 24$
$(4, 1)$	orbit$((1\ 2\ 3\ 4))$	$\binom{5}{4} \times 3! = 30$
$(3, 2)$	orbit$((1\ 2\ 3)(4\ 5))$	$\binom{5}{3} \times 2! = 20$
$(3, 1, 1)$	orbit$((1\ 2\ 3))$	$\binom{5}{3} \times 2! = 20$
$(2, 2, 1)$	orbit$((1\ 2)(3\ 4))$	$\binom{5}{2} \times \binom{3}{2} \times \frac{1}{2} = 15$
$(2, 1, 1, 1)$	orbit$((1\ 2))$	$\binom{5}{2} = 10$
$(1, 1, 1, 1, 1)$	orbit(1)	1
		$\Sigma = 120$

From this table we can draw an interesting conclusion: "*The center of S_5 is the trivial group.*" Recall that for a group G, its center of defined by $Z(G) := \{x \in G : gx = xg$ forall $g \in G$. We note that $x \in Z(G)$ if and only if the conjugacy class of x is singleton set, containing x alone. Now, look at the table and conclude that $Z(S_5) = \{1\}$.

Can you find the center of S_n, for $n \neq 5$? □

Lecture 19

Platonic Solids

March 21, 2025

Let us start with a question: *How many convex regular polygons can be inscribed in a circle?*

The answer is obvious, *infinitely many*. The groups of rotations of these polygons are cyclic groups. These are precisely all the subgroups of $\mathrm{SO}_2(\mathbb{R})$. The proof of this is analogous to the proof that all subgroups of $(\mathbb{Z}, +)$ are cyclic. Note that $\mathrm{SO}_2(\mathbb{R})$ is the group of rotational symmetries of a circle $S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$.

We try to imitate the above for *three-dimensional* situation. The group of rotational symmetries of a (unit) sphere $S^2 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ is $\mathrm{SO}_3(\mathbb{R})$. There are multiple *axes of rotation* in S^2. If we choose the axis as Z-axis and rotate the sphere in the anti-clockwise direction (as seen from the 'positive' part the of Z-axis), the corresponding rotation matrix will be:

$$R_{\theta,Z} = \begin{pmatrix} \cos\,\theta & -\sin\,\theta & 0 \\ \sin\,\theta & \cos\,\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Similarly for other axes we have:

$$R_{\theta,X} = \begin{pmatrix} 0 & 0 & 1 \\ \cos\,\theta & -\sin\,\theta & 0 \\ \sin\,\theta & \cos\,\theta & 0 \end{pmatrix} \quad \text{and} \quad R_{\theta,Y} = \begin{pmatrix} \cos\,\theta & 0 & \sin\,\theta \\ 0 & 1 & 0 \\ -\sin\,\theta & 0 & \cos\,\theta \end{pmatrix}$$

The group $\mathrm{SO}_3(\mathbb{R})$ is not commutative: $R_{\frac{\pi}{2},X} R_{\frac{\pi}{2},Y} \neq R_{\frac{\pi}{2},Y} R_{\frac{\pi}{2},X}$.

How many regular convex polyhedra can be inscribed inside a sphere? First, let us understand the meaning of regular convex polyhedron.

A convex polyhedron is said to be *regular* if:

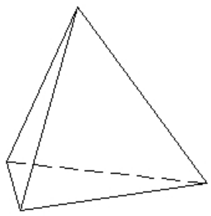
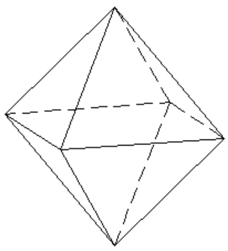
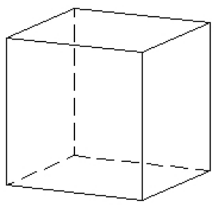
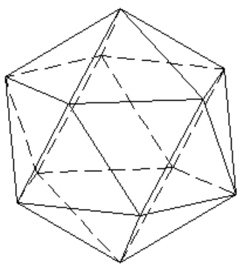
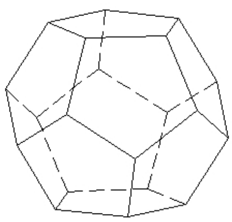
1. Its faces are convex regular n-gons (for some $n \in \mathbb{N}$, $n \geq 3$) and any two faces are congruent.

2. Only possible intersections of faces are at edges.

3. At each vertex the same number (say m) of edges meet.

It is clear that n and m are at least 3. Regular convex polyhedra are also called *Platonic solids*. Cube is an example of a Platonic solid. For a cube $n = 4$, as each face is a square (regular 4-gon), and $m = 3$ as at each vertex three faces meet.

For a Platonic solid the pair $\{n, m\}$ is called its *Schläfli symbol*. For a tetrahedron the Schläfli symbol is $\{3, 3\}$. Here is a list of Platonic solids:

Shape	**Name**	**Schläfli symbol**
	Tetrahedron	$\{4, 4\}$
	Octahedron	$\{3, 4\}$
	Cube	$\{4, 3\}$
	Icosahedron	$\{3, 5\}$
	Dodecahedron	$\{5, 3\}$

What is remarkable is that this is the complete list of Platonic solids. To prove it we shall use the following result from topology/graph theory.

Theorem 19.1 (Euler)**.** *If a polyhedron has v vertices, f faces and e edges then $v - e + f = 2$.*

Proof. The proof is not very difficult. A proof using induction on v and another using dual graphs can be found in the book Basic Topology by M.A. Armstrong. If you have studied Algebraic Topology then you should be able to relate it with the Euler characteristic of the 2-sphere. □ □

Theorem 19.2 (Euclid)**.** *There are only five Platonic solids: tetrahedron, cube, octahedron, icosahedron and dodecahedron.*

Proof. Let us count the number of edges in two different ways:

1. Since each face consists of n edges and there are f faces, the total number of edges counted twice (as each edge belongs to two faces) is nf. Therefore we conclude that $nf = 2e$.

2. At each vertex m edges meet. Thus the total number of edges counted twice (as each edge corresponds to two vertices) is mv. Thus $mv = 2e$.

We plug in this information in the Euler's formula to get:

$$v - e + f = \frac{2e}{m} - e + \frac{2e}{n} = 2.$$

Therefore

$$\frac{1}{m} + \frac{1}{n} = \frac{1}{2} + \frac{1}{e} > \frac{1}{2}.$$

Since $n \geq 3$ and $m \geq 3$, in view of above inequality, the only possibilities are:

1. $n = m = 3$: In this case $e = 6, v = f = 4$ and *tetrahedron* is the corresponding Platonic solid.

2. $n = 3, m = 4$: In this case $e = 12, v = 6, f = 8$ and *octahedron* is the corresponding Platonic solid.

3. $n = 4, m = 3$: In this case $e = 12, v = 8, f = 6$ and *cube* is the corresponding Platonic solid.

4. $n = 3, m = 5$: In this case $e = 30, v = 12, f = 20$ and *icosahedron* is the corresponding Platonic solid.

5. $n = 5, m = 3$: In this case $e = 30, v = 20, f = 12$ and *dodecahedron* is the corresponding Platonic solid.

□

Each of the Platonic solids can be inscribed in a sphere. Any rotational symmetry of these solids gives a rotational symmetry of the sphere and hence the rotational symmetry groups of these solids are subgroups of $\mathrm{SO}_3(\mathbb{R})$.

Exercise 19.3 Prove that groups of rotational symmetries of Platonic solids are finite. To show this, it is enough to show that there are finitely many axes of rotations and about each axis only finitely many rotations give a symmetry.

Lecture 20

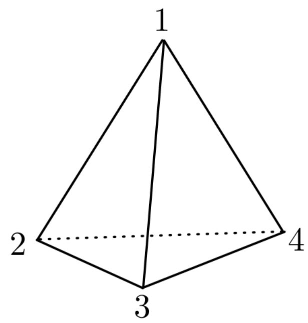
Revisiting symmetries of Platonic Solids

March 24, 2025

We now revisit the symmetry groups of Platonic solids.

20.1. Tetrahedron

Imagine a tetrahedron that is kept on a plane. Enumerate the four vertices as $1, 2, 3$ and 4 with the vertex 1 on the top.



There are two types of axes in the tetrahedron:

1. *The ones for which one end of the axis is a vertex*: For these rotations the other end of the axis has to be there at the mid of the opposite face. For the vertex 1 there are two non-trivial rotations. We call them $(2\ 3\ 4)$ and $(2\ 4\ 3)$. Other similar rotations are: $(1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4)$ and $(1\ 4\ 2)$.

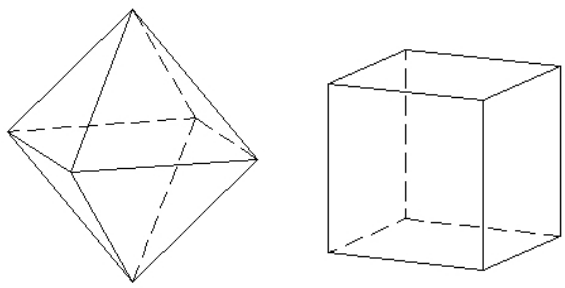
2. *The ones for which the axes is the line joining mid points of two edges having no common vertex*: Consider the mid points of the edge $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$. Taking the line joining these two mid points gives an axis of rotation of the tetrahedron. This gives one non-trivial rotation which we denote by $(1\ 2)(3\ 4)$. Other such rotations are: $(1\ 3)(2\ 4)$ and $(1\ 4)(2\ 3)$.

Trivial rotation, the identity, is still missing. Including this too in the list we have the rotation group of Tetrahedron as:

$$\begin{aligned}\mathrm{rot}(\mathbb{T}) = \{&1, (2\,3\,4), (2\,4\,3), (1\,2\,3), (1\,3\,2), (1\,3\,4), (1\,4\,3),\\ &(1\,2\,4), (1\,4\,2), (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}\end{aligned}$$

It is important to establish the composition of symmetries of a tetrahedron indeed goes exactly on the same lines as the composition of symmetries of the alternating group A_4

20.2. Cube and Octahedron



The symmetry groups of cube and octahedron are same. This is because they are dual in the sense of 'reversing' the roles of faces and vertices. We start with identifying axes for the rotational symmetry of the cube. There are three types of axes:

1. *The ones whose axes pass through the mid points of opposite faces*: There are three such axes. Further for each such axis there are three non-trivial rotations (by $90°$, $180°$ and $270°$). Thus the total number of rotations arising this way is: $3 \times 3 = 9$.

2. *The ones whose axes pass through the mid points of opposite edges*: There are six such axes. Further for each such axis there is only one non-trivial rotation (by $180°$). Thus the total number of rotations arising this way is: $6 \times 1 = 6$.

3. *The ones whose axes pass through opposite vertices*: There are four such axes. Further for each such axis there are two non-trivial rotations (by $120°$ and $240°$). Thus the total number of rotations arising this way is: $4 \times 2 = 8$.

In the end we put identity, the trivial rotation, to make the total count $9 + 6 + 8 + 1 = 24$.

Let $\langle s, t\, :\, s^2, t^3, (st)^4$ denote the group with generating set $\{s, t\}$ and relations $s^2 = 1, t^3 = 1, (st)^4 = 1$. Let s' denote a non-trivial rotation about an axes of type 2 above and t' denote a rotation of type 3 then the product $s't'$ is a $90°$ rotation of type 1.

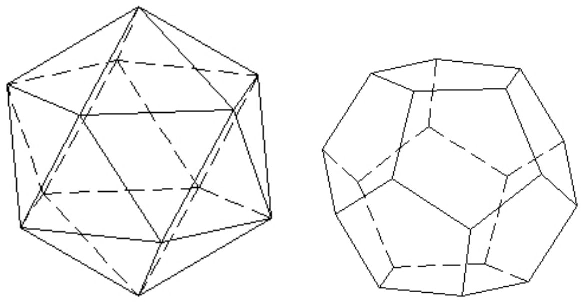
This suggests that $s \to s'$ and $t \to t'$ extends to a surjective homomorphism

$$\langle s, t : s^2, t^3, (st)^4\rangle \to \mathrm{rot}(\mathcal{C}).$$

Since $\langle s, t : s^2, t^3, (st)^4\rangle \simeq S_4$ and $\mathrm{rot}(\mathcal{C})$ is of size 24, the above homomorphism is an isomorphism and we have $\mathrm{rot}(\mathcal{C}) \simeq S_4$.

20.3. Icosahedron and Dodecahedron

Since icosahedron and dodecahedron are dual, their symmetry groups are same. As before, we start with identifying axes for the rotational symmetry of icosahedron. There



are three types of axes:

1. *The ones whose axes pass through the mid points of opposite faces*: There are ten such axes. Further for each such axis there are two non-trivial rotations (by $120°$ and $240°$). Thus the total number of rotations arising this way is: $10 \times 2 = 20$.

2. *The ones whose axes pass through the mid points of opposite edges*: There are fifteen such axes. Further for each such axis there is only one non-trivial rotation (by $180°$). Thus the total number of rotations arising this way is: $15 \times 1 = 15$.

3. *The ones whose axes pass through opposite vertices*: There are six such axes. Further for each such axis there are four non-trivial rotations (by $120°$ and $240°$). Thus the total number of rotations arising this way is: $6 \times 4 = 24$.

Along with the identity rotation the total count is $20 + 15 + 24 + 1 = 60$. As in the previous cases, we establish a surjective homomorphism

$$\langle s, t : s^2, t^3, (st)^5 \rangle \to \mathrm{rot}(\mathcal{I})$$

and use the isomorphism $\langle s, t : s^2, t^3, (st)^5 \rangle \simeq A_5$ and size count of $\mathrm{rot}(\mathcal{I})$ to prove that $\mathrm{rot}(\mathcal{I}) \simeq A_5$.

There is a another way to see the isomorphism $\mathrm{rot}(\mathcal{I}) \simeq A_5$.

20.3.3.1. $\mathrm{rot}(\mathcal{I})$ *is simple*

A group G is called *simple* if is the only normal subgroup it admits are $\{1\}$ and G itself. Thus, a simple group does not admit nontrivial proper subgroups. Here we show that $\mathrm{rot}(\mathcal{I})$, the group of rotational symmetries of an icosahedron, is simple.

The distribution of 60 elements of this group, according to their orders is:

Order	Number of elements	Axis
1	1	–
2	15	Mid points of edges
3	20	Mid points of faces
5	24	Vertices

Thus we have a partition $60 = 1 + 15 + 20 + 24$. Since any two conjugate elements are have same order, the class equation of $\text{rot}(\mathcal{I})$ is a refinement of the above partition. Observe that $\text{rot}(\mathcal{I})$ acts transitively on the set of edges of the icosahedron. Thus all the groups of order 2 are conjugate to each other. The same is true for groups of order 3 and 5 - the action of $\text{rot}(\mathcal{I})$ on the set of faces and vertices is also transitive. From this it is evident that all 15 elements of order 2 are conjugate to each other.

We claim that all elements of order 3 are also conjugate. Any element of order 3 lies in some subgroup of the form $\text{stab}(F) = \{1, g, g^2\}$ for some face F. If r is the rotation by $180°$ about an axis orthogonal to the axis passing through the mid point of F then $rgr^{-1} = g^2$. Further, since $\text{rot}(\mathcal{I})$ acts transitively on the set of faces of an icosahedron, the groups $\text{stab}(F)$ and $\text{stab}(F')$ are conjugate. From this we conclude that any two elements of order three are conjugate.

Now finally, we see that elements of order 5 split into two conjugacy classes. By the argument as in the case of order 3 elements, rotations by $72°$ and $288°$ are conjugate and those by $144°$ and $216°$ are conjugate. Also, the rotations by $72°$ are not conjugate to the ones by $144°$, else all 24 elements of order 5 will form a conjugacy classes which is a contradiction in because $24 \nmid 60$.

The class equation, therefore, of the group $\text{rot}(\mathcal{I})$ is:

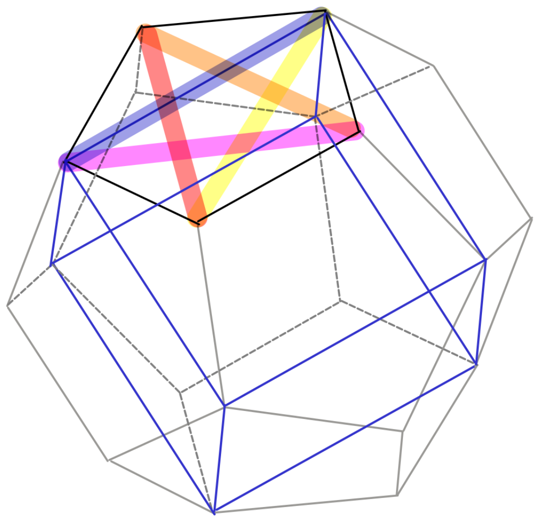
$$60 = 1 + 15 + 20 + 12 + 12.$$

If there were normal subgroups in $\text{rot}(\mathcal{I})$, they would be disjoint unions of conjugacy classes along with the identity element. However none of the partial sums (with 1 as a term) in the above class equation divides 60. This shows that the group $\text{rot}(\mathcal{I})$ does not have normal subgroups.

20.3.3.2. $\text{rot}(\mathcal{I}) \simeq A_5$

We first show that there is a homomorphism from $\text{rot}(\mathcal{I}) \to S_5$. For this it is enough to give an action of $\text{rot}(\mathcal{I})$ on a set of five elements. This set is actually the set of five cubes inscribed in a Dodecahedron. One of the five such cubes is shown in the following figure:

Since $\text{rot}(\mathcal{I})$ is simple, the homomorphism $\text{rot}(\mathcal{I}) \to S_5$ is injective. Let G be the image of this homomorphism. We restrict the signature homomorphism $S_5 \to \{-1, 1\}$ to the



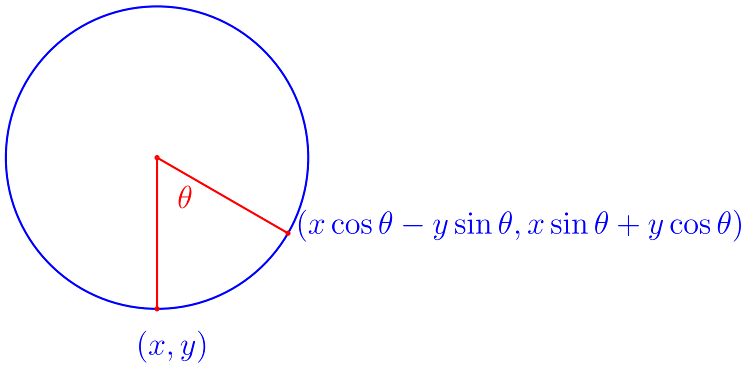
subgroup G. If $G \to \{-1, 1\}$ is surjective then its kernel will be a normal subgroup of order 30, which can not be the case as G is simple, being isomorphic to $\mathrm{rot}(\mathcal{I})$. Thus $G \subseteq A_5$. Counting the number of elements gives: $\mathrm{rot}(\mathcal{I}) \simeq G = A_5$. $\square$

Lecture 21

Quaternions and rotations

April 02, 2025

We recall rotations in $\mathbb{R}^2$.



Since

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix},$$

these rotations can be expressed through the rotation matrix:

$$R_\theta := \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

We can also deploy complex multiplication (multiplication of complex numbers) to understand rotations in $\mathbb{R}^3$. Under the identification $(a, b) \leftrightarrow (a + bi)$, the operation:

$$(a, b) \circ (x, y) = (ax - by, ay + bx)$$

is precisely the multiplication of complex numbers. Further, multiplying (x, y) by $(a, b) := (\cos\theta, \sin\theta)$ gives rotation of (x, y) by angle θ in anticlockwise direction. In particular, multiplication by i, which corresponds to $(0, 1)$, gives rotation by $\pi/2$. To understand rotations in $\mathbb{R}^3$, we wish to have a system similar to complex multiplications.

Recall the rotation matrices in $\mathbb{R}^3$, about three coordinate axes:

$$R_{\theta,x} := \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \quad R_{\theta,y} := \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$R_{\theta,z} := \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

How to understand rotations about other (infinitely many) axes? There are classic approaches such as Euler's Rotation Theorem, that states that any rotation in $\mathbb{R}^3$ is a product $R_{\psi,z}R_{\theta,x}R_{\phi,z}$ for suitable anglesd (ψ, θ, ϕ). However, this approach poses the some problems. First, it is difficult to determine even the axis of rotation, and second, for a desired rotation, it is hard to compute the Euler angles. That's where quaternions become a useful tool.

21.1. Quaternionic Multiplication

Quaternions were not discovered with the intention to use them to understand rotations, but out of curiosity to find a "complex-multiplication" like structure on $\mathbb{R}^3$. The discovery of quaternions marks a significant moment in the history of mathematics. In 1843, Irish mathematician Sir William Rowan Hamilton introduced quaternions as an extension of complex numbers. Hamilton had long been searching for a way to extend the complex multiplication $(a, b) \circ (x, y) = (ax - by, ay + bx)$ to three dimensions. He repeatedly encountered obstacles when trying to preserve multiplication rules. The breakthrough came when he was walking near Broom Bridge on the Royal Canal in Dublin with his wife on October 16, 1843. In an 'aha! moment', he realized that he would need a 4-dimensional number system to accomplish his goal. Legend is, he immediately carved the fundamental quaternion identity

$$i^2 = j^2 = k^2 = ijk = -1$$

into the stone of the bridge.

Quaternions introduced a new kind of number system involving one real axis and three 'imaginary' axes i, j, k, that span the four dimension vector space $\mathbb{R}^4$. A typical element in this vector space is $a + bi + cj + dk \in \mathbb{H}$. Unlike real or complex numbers, quaternion multiplication is non-commutative: $ij \neq ji$. This was a radical idea at the time that challenged traditional algebraic thinking!

For $q = a + bi + cj + dk$ and $q' = x + yi + zj + wk$ in $\mathbb{R}^4$, the conditions $i^2 = j^2 = k^2 = ijk = -1$, along with distributivity, enforce

$$\begin{aligned} q * q' = &(ax - by - cz - dw) \\ &+ (ay + bx + cw - dz)i \\ &+ (az - bw + cx + dy)j \\ &+ (aw + bz - cy + dx)k \end{aligned}$$

The space $\mathbb{R}^4$ equipped with $*$ is called *quaternion algebra*. It is denoted by $\mathbb{H} := (\mathbb{R}^4, *)$. An element of quaternion algebra is called a *quaternion*.

21.2. Rotations via Quaternions

We write $a + bi + cj + dk =: [\alpha, \vec{u}]$ for a typical element of $\mathbb{H}$. In this notation, $\alpha \in \mathbb{R}$ is called the *scalar* part and $\vec{u} := bi + cj + dk \in \mathbb{R}^3$ is called the *vector* part of the given quaternion.

For $q := [\alpha, \vec{u}] \in \mathbb{H}$, we define its conjugate to be the quaternion $\overline{q} := [\alpha, -\vec{u}]$. We check that $||q||^2 := q\overline{q} = \alpha^2 + ||\vec{u}||^2$. Here, $||\vec{u}||$ is the length of the three dimensional vector $\vec{u}$. Therefore, if $q \neq [0, \vec{0}]$, then

$$q^{-1} = \frac{\overline{q}}{\alpha^2 + ||\vec{u}||^2}$$

In other words, every nonzero element in a quaternion algebra is invertible.

Through a simple calculation, we obtain the following amazing expression to describe quaternion multiplication:

$$[\alpha, \vec{u}] * [\beta, \vec{v}] = [\alpha\beta - \vec{u}.\vec{v}, (\alpha\vec{v} + \beta\vec{u}) + \vec{u} \times \vec{v}]$$

Let us have $\theta \in \mathbb{R}$ with an intention to signify an angle, and $\vec{u} = (u_x, u_y, u_z) \in \mathbb{R}^3$ be a unit vector, that we intend to use as an axis. Consider the quaternion

$$q = \left[\cos\frac{1}{2}\theta, (u_x, u_y, u_z)\sin\frac{1}{2}\theta\right]$$

We observe that $||q|| = 1$. It is not difficult to show that all quaternions with $||q|| = 1$ are of this form.

Let $\vec{p} = (p_x, p_y, p_z) \in \mathbb{R}^3$. Then

$$\begin{aligned} q * [0, \vec{p}] * q^{-1} &= \left[\cos\frac{1}{2}\theta, (u_x, u_y, u_z)\sin\frac{1}{2}\theta\right] * [0, \vec{p}] * \left[\cos\frac{1}{2}\theta, -(u_x, u_y, u_z)\sin\frac{1}{2}\theta\right] \\ &= [0, R_{\theta,\vec{u}}(\vec{p})] \end{aligned}$$

Where $R_{\theta,\vec{u}}(\vec{p})$ is the rotation of $\vec{p}$ by angle θ about axis $\vec{u}$. Quaternions are widely used for computer animation, spacecraft control, Robotics, and computer vision.

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